Experiment No-01

Name of the experiment:

Determination of the molecular weight of a substance by cryoscopic method.

Data table:

Data table for the determination of freezing point of pure cyclohexane with time:

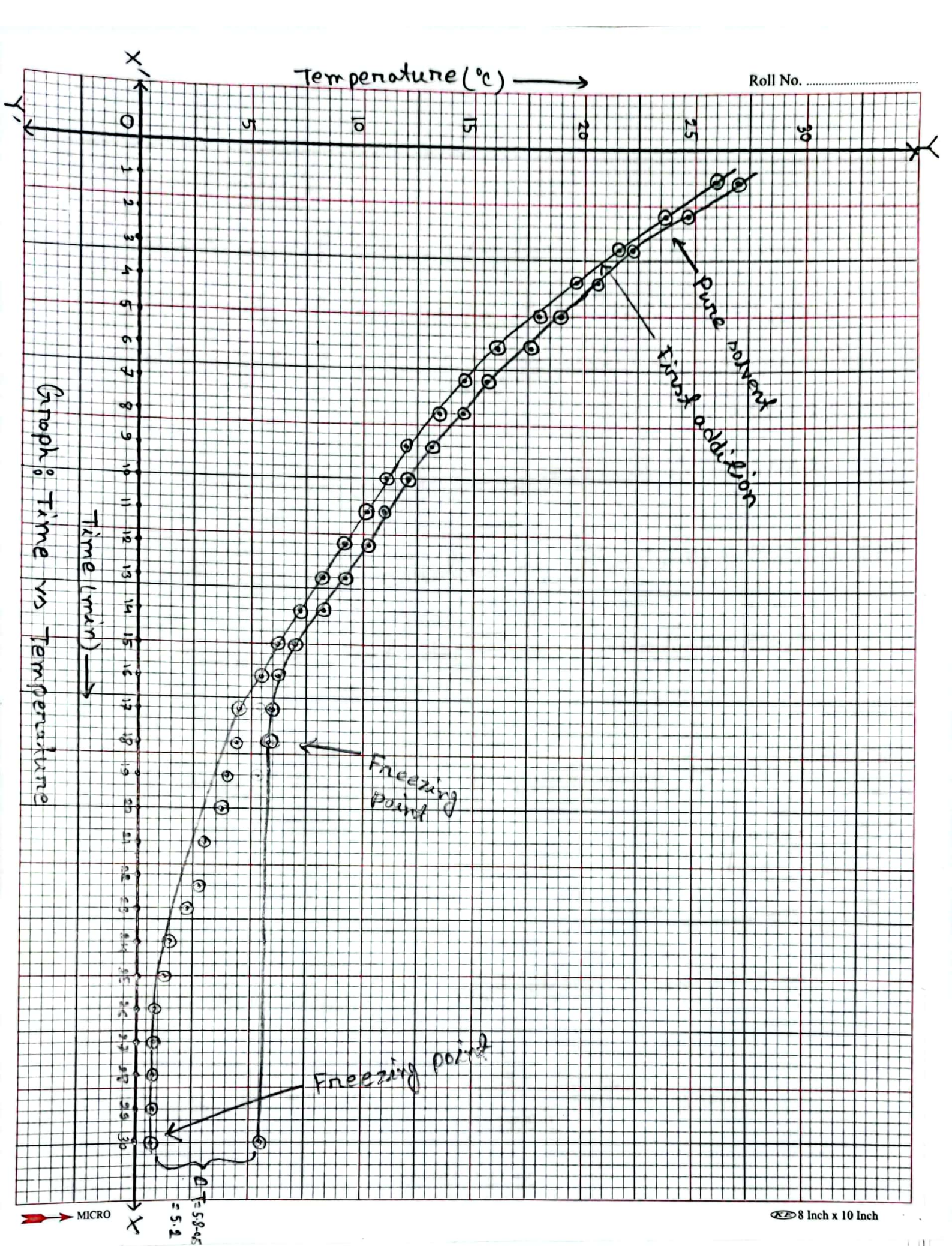
Time (min)	Temperature (°C)	Time (min)	Temperature (°C)
0.0	27.7	10.0	12.1
0.5	27.5	10.5	11.7
1.0	26.5	11.0	11.0
1.5	25.7	11.5	10.3
2.0	24.6	12.0	10.1
2.5	23.6	12.5	9.4
3.0	22.4	13.0	8.9
3.5	21.6	13.5	8.4
4.0	20.9	14.0	8.1
4.5	20.0	14.5	7.5
5.0	19.0	15.0	7.0
5.5	18.0	15.5	6.8
6.0	17.6	16.0	6.3
6.5	16.8	16.5	6.2
7.0	15.9	17.0	5.8
7.5	15.4	17.5	5.8
8.0	14.7	18.0	5.8
8.5	14.0	18.5	5.8
9.0	13.4	19.0	5.8
9.5	12.9	19.5	5.8

Data table:

Data ~~poi~~ for the determination of the freezing point of the mixture of cyclohexane and 0.52g Napthalene.

Time (min)	Temperature (°c)	Time (min)	Temperature (°c)
0.00	27.1	10.0	11.3
0.5	26.8	10.5	10.8
1.0	26.0	11.0	10.3
1.5	24.8	11.5	9.7
2.0	23.7	12.0	9.1
2.5	22.7	12.5	8.8
3.0	21.6	13.0	8.2
3.5	20.7	13.5	7.6
4.0	19.8	14.0	7.3
4.5	18.8	14.5	7.0
5.0	18.0	15.0	6.4
5.5	17.2	15.5	6.1
6.0	16.4	16.0	5.7
6.5	15.8	16.5	5.4
7.0	14.9	17.0	5.0
7.5	14.5	17.5	4.7
8.0	13.8	18.0	4.7
8.5	13.1	18.5	4.3
9.0	12.6	19.0	4.0
9.5	12.1	19.5	3.7

Time (min)	temperature(°C)	Time (min)	Temperature (°C)
20.0	3.7	31.0	0.6
20.5	3.3	31.5	0.6
21.0	3.0	32.0	
21.5	3.0	32.5	
22.0	2.7	33.0	
22.5	2.7	33.5	
23.0	2.3	34.0	
23.5	1.9	34.5	
24.0	1.5	35.0	
24.5	1.5	35.5	
25.0	1.2	36.0	
25.5	1.2	36.5	
26.0	0.8	37.0	
26.5	0.8	37.5	
27.0	0.7	38.0	
27.5	0.7	38.5	
28.0	0.6	39.0	
28.5	0.6	39.5	
29.0	0.6	40.0	
29.5	0.6		
30.0	0.6		
30.5	0.6		


Temperature (°C) →
Roll No. .....
X'
Y'
O
5
10
15
20
25
30
Time (min) →
Graph of Time vs Temperature
Pure solvent
First addition
Freezing point
Freezing point
ΔT = 5.8-5.5 = 5.2
MICRO
8 Inch x 10 Inch

Calculation:

1. Volume of solvent (cyclohexane) = 23.00 mL
2. Density of solvent (cyclohexane) = 0.773 g/mL
3. Weight of solvent (cyclohexane), $w_2 = (23 \times 0.77)$
   $= 17.71$ g
4. Weight of solute (naphthalene), $w_1 = 0.56$ g
5. Cryoscopic constant of solvent, $K_f = 20.2$ deg.kg/mole
6. Freezing point depression, $\Delta T = (5.8 - 0.6)^\circ C$
   $= 5.2^\circ C$
7. Molecular weight of napthalene, $M = \frac{1000 \times K_f \times w_1}{w_2 \times \Delta T}$
   $= \frac{1000 \times 20.2 \times 0.56}{17.71 \times 5.2}$
   $= 122.83$ g/mol
8. Actual weight of napthalene = 128.17 g/mol
9. % of error $= \frac{128.17 - 122.83}{128.17} \times 100\%$
   $= 4.16\ \%$

Result:

1. Molecular weight of napthalene = 122.83 g/mol
2. % of error = 4.16 %

Discussion:

1. The freezing point of solution is less than that of the pure solvent this is thermodynamic factor.
2. In this experiment, cyclohexane was used as solvent, and napthalene is used as solute.
3. Temperature reading was taken after 30 second and a digital thermometer was used to take the temperature reading.
4. The solution was constant stirred with glass rod.
5. Finally we have calculate the data sincerely and get a nice result with a little error of 4.16 %.

Date: 07.08.23
Group no-08
Roll no-205

Experiment NO-02

Name of the experiment:

Study of the adsorption of acetic acid on charcoal and examine the validity of Freundlich's adsorption isotherm and Langmuirs adsorption isotherm.

Data:

To prepare 0.5M CH_3COOH solution:

① The molar mass of $CH_3COOH = 60\ g\ mol^{-1}$

② For 100mL of standard 0.5M CH_3COOH sol^n the weight of the acetic acid to be taken = 3g

③ weight of the CH_3COOH is taken = 3.01g

Table: Data table for the titration of NaOH against 0.5M CH_3COOH.

No of obs.	Vol^m of CH_3COOH (mL)	Vol^m of sodium hydroxide			Mean (mL)
		IBR (mL)	FBR (mL)	Diff (mL)	
1	5	21.0	43.2	22.2	22.2
2	5	0.0	22.2	22.2	
3	5	22.2	44.4	22.2	

Bottle-1:

⇒ Bottle-1 contain 0.5N acetic acid 50mL, distilled water 0mL, charcoal 1gm.

⇒ volume of NaOH for 10mL acetic acid before adsorption ($V_1 = 50mL$)

Table for the titration of 0.1N NaOH against after adsorbed acetic acid:

No of obs	volm of CH_3COOH (mL)	volm of 0.1N NaOH soln			mean (mL)
		IBR (mL)	FBR (mL)	Diff (mL)	
1	10	0.0	42.2	42.2	42.2
2	10	0.0	42.2	42.2	
3	10	0.0	42.2	42.2	

Bottle-2:

⇒ Bottle-2 contain 40mL of 0.5N CH_3COOH, 10mL of Distilled water, 1g of charcoal.

⇒ Volume of the NaOH for 10mL of acetic acid before adsorption ($V_1 = 40mL$)

Data table for the titration of 0.1 N NaOH against after adsorbed acetic acid:

No of obs	vol^m of CH_3COOH (mL)	vol^m of 0.1 N NaOH soln			Mean (mL)
		IBR (mL)	FBR (mL)	Diff (mL)	
1	10	10.0	42.0	32.0	
2	10	0.0	32.0	32.0	32.0
3	10	0.0	32.0	32.0	

Bottle-3:

⇒ This bottle contain 30mL of 0.5 N CH_3COOH, 20mL of distilled water, 1g of charcoal.

⇒ volume of the 0.1 N NaOH for 10mL of acetic acid solution before adsorption (V_1 = 30 mL)

Data table for the titration of 0.1 N NaOH against after adsorbed acetic acid solution:

No of obs	vol^m of CH_3COOH (mL)	vol^m of 0.1 N NaOH soln			Mean (mL)
		IBR (mL)	FBR (mL)	Diff (mL)	
1	10.0	0.0	24.5	24.5	
2	10.0	15.3	39.8	24.5	24.5
3	10.0	0.0	24.5	24.5	

Bottle-4:

⇒ This bottle contain 25mL of 0.5 N CH_3COOH, 25mL of distilled water, 1g of charcoal.

⇒ volume of the 0.1 N NaOH for 10mL of acetic acid solution before adsorption ($V_1 = 25mL$)

⇒ Data table for the titration of 0.1N NaOH against after adsorbed acetic acid solution:

No of obs	Volm of CH_3COOH (mL)	Volm of 0.1 N NaOH soln			Mean (mL)
		IBR(mL)	FBR(mL)	Diff(mL)	
1	10	0.0	19.3	19.3	19.3
2	10	20.0	39.3	19.3	
3	10	10.0	29.3	19.3	

Bottle-5:

⇒ This bottle contain 20mL of 0.5 M CH_3COOH, 30 mL of distilled water, 1g of charcoal.

⇒ volume of the 0.1 N NaOH for 10mL of acetic acid solution before adsorption

($V_1 = 20mL$)

Data table for the titration of 0.1 N NaOH against after adsorbed acetic acid solution:

No of obs.	Volm of CH_3COOH (mL)	Volm of 0.1 N NaOH soln			Mean (mL)
		IBR (mL)	FBR (mL)	Diff. (mL)	
1	10	0.0	15.6	15.6	15.6
2	10	6.6	22.2	15.6	
3	10	10.0	25.6	15.6	

Bottle-6:

⇒ This bottle contain 10mL of 0.5M CH_3COOH, 40mL of distilled water, 1g of ~~ethy~~ charcoal.

⇒ Volume of the $\overset{0.1N}{\wedge}$ NaOH for 10mL of acetic acid solution before adsorption (V_1 = 10mL)

⇒ Data table for the titration of 0.1 N NaOH against after adsorbed acetic acid solution:

No of obs.	Volm of CH_3COOH (mL)	Volm of 0.N NaOH soln			Mean (mL)
		IBR (mL)	FBR (mL)	Diff (mL)	
1	10	0.0	7.5	7.5	7.5
2	10	10.0	17.5	7.5	
3	10	20.0	27.5	7.5	

Calculation :

1. Strength of CH_3COOH = $\frac{\text{weight of } CH_3COOH \text{ taken}}{\text{weight of } CH_3COOH \text{ to be taken}} \times$ Strength of sol^n to be made

$$= \frac{3.01}{3.00} \times 0.5$$

$$= 0.5\ N\ CH_3COOH$$

2. We know,

$$S_1V_1 = S_2V_2$$

$$\Rightarrow S_2 = \frac{S_1V_1}{V_2}$$

$$= \frac{0.2 \times 5}{22.2}$$

$$= 0.113$$

$$= 0.1\ N \text{ (approx)}$$

Here,
volume of CH_3COOH, $V_1 = 5\ mL$
conc. of CH_3COOH, $S_1 = 0.5\ N$
volume of NaOH, $V_2 = 22.2\ mL$
conc. of NaOH, $S_2 = ?$

Therefore, the strength of NaOH is 0.1 N

i) Normality of NaOH = 0.1 N

ii) Eq. weight of acetic acid = 60 g/eq.

iii) weight of charcoal, m = 1 g

iv) $(V_1 - V_2)$ indicate the adsorbed acetic acid.

v) The factor 5 comes in x/m calculation because 10mL is taken for titration whereas set was made at 50mL of 1g of charcoal.

vi) A multiplication factor of 1000 is used in Ce calculation to have positive log ce value.

Bottle No. 1:

volume for NaOH for 10mL acetic acid before adsorption, V_1 = 50mL

volume of NaOH for 10mL acetic acid after adsorption V_2 = 42.2 mL

i) Equilibrium concentration, $Ce = \frac{V_2 \times N_{NaOH} \times 1000}{10}$

$= \frac{42.2 \times 0.1 \times 1000}{10}$

= 422 milli eq/L

We know,

$$\frac{x}{m} = \frac{5(V_1 - V_2)}{1} \times 0.1 \times 60 \text{ mg of acetic acid per g of charcoal.}$$

$$= \frac{5(50 - 42.2)}{1} \times 0.1 \times 60$$

$$= 234 \text{ mg.}$$

Bottle - 2 :

$$V_1 = 40 \text{ mL}$$

$$V_2 = 32 \text{ mL}$$

$$\therefore Ce = \frac{V_2 \times 0.1 \times 1000}{10}$$

$$= \frac{32 \times 0.1 \times 1000}{10}$$

$$= 320 \text{ milli eq/L}$$

$$\therefore \frac{x}{m} = \frac{5(V_1 - V_2) \times 0.1 \times 60}{1}$$

$$= \frac{5(40 - 32) \times 0.1 \times 60}{1}$$

$$= 240 \text{ mg}$$

Bottle - 3 :

$$V_1 = 30 \text{ mL}$$

$$V_2 = 24.5 \text{ mL}$$

$$\therefore Ce = \frac{V_2 \times 0.1 \times 1000}{10}$$

$$= \frac{24.5 \times 0.1 \times 1000}{10}$$

$$= 245 \text{ milli eq/L}$$

$$\therefore \frac{x}{m} = \frac{5 (V_1 - V_2) \times 0.1 \times 60}{1}$$

$$= \frac{5(30 - 24.5) \times 0.1 \times 60}{1}$$

$$= 165 \text{ mg}$$

bottle-4:

$V_1 = 25 \text{ mL}$

$V_2 = 19.3 \text{ mL}$

$$\therefore Ce = \frac{V_2 \times 0.1 \times 1000}{10}$$

$$= \frac{19.3 \times 0.1 \times 1000}{10}$$

$$= 193 \text{ milli eq/L}$$

$$\therefore \frac{x}{m} = \frac{5 (V_1 - V_2) \times 0.1 \times 60}{1}$$

$$= \frac{5 (25 - 19.3) \times 0.1 \times 60}{1}$$

$$= 171 \text{ mg}$$

Bottle- 5:

$V_1 = 20$ mL

$V_2 = 15.6$ mL

$$\therefore Ce = \frac{V_2 \times 0.1 \times 1000}{10}$$

$$= \frac{15.6 \times 0.1 \times 1000}{10}$$

$$= 156 \text{ mili eq/L}$$

$$\therefore \frac{x}{m} = \frac{5(V_1 - V_2) \times 0.1 \times 60}{1}$$

$$= \frac{5 \times (20 - 15.6) \times 0.1 \times 60}{1}$$

$$= 132 \text{ mg}$$

Bottle-6:

$V_1 = 10$ mL

$V_2 = 7.5$ mL

$$\therefore Ce = \frac{V_2 \times 0.1 \times 1000}{10}$$

$$= \frac{7.5 \times 0.1 \times 1000}{10}$$

$$= 75 \text{ mili eq/L}$$

$$\therefore \frac{x}{m} = \frac{5(V_1 - V_2) \times 0.1 \times 60}{1}$$

$$= \frac{5(10 - 7.5) \times 0.1 \times 60}{1}$$

$$= 75 \text{ mg}$$

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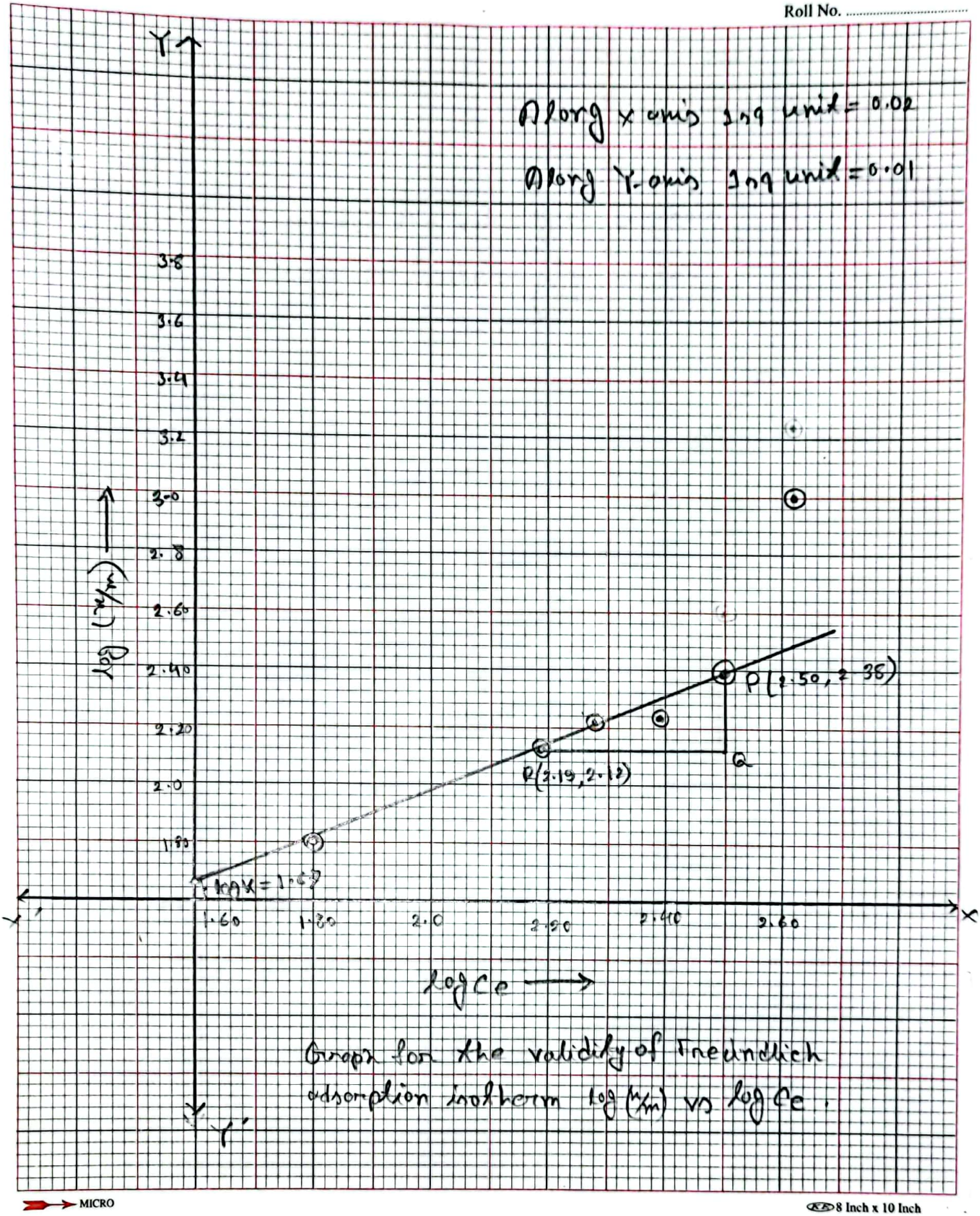


Graph for the validity of Freundlich adsorption isotherm $\log(x/m)$ vs $\log C_e$.

MICRO

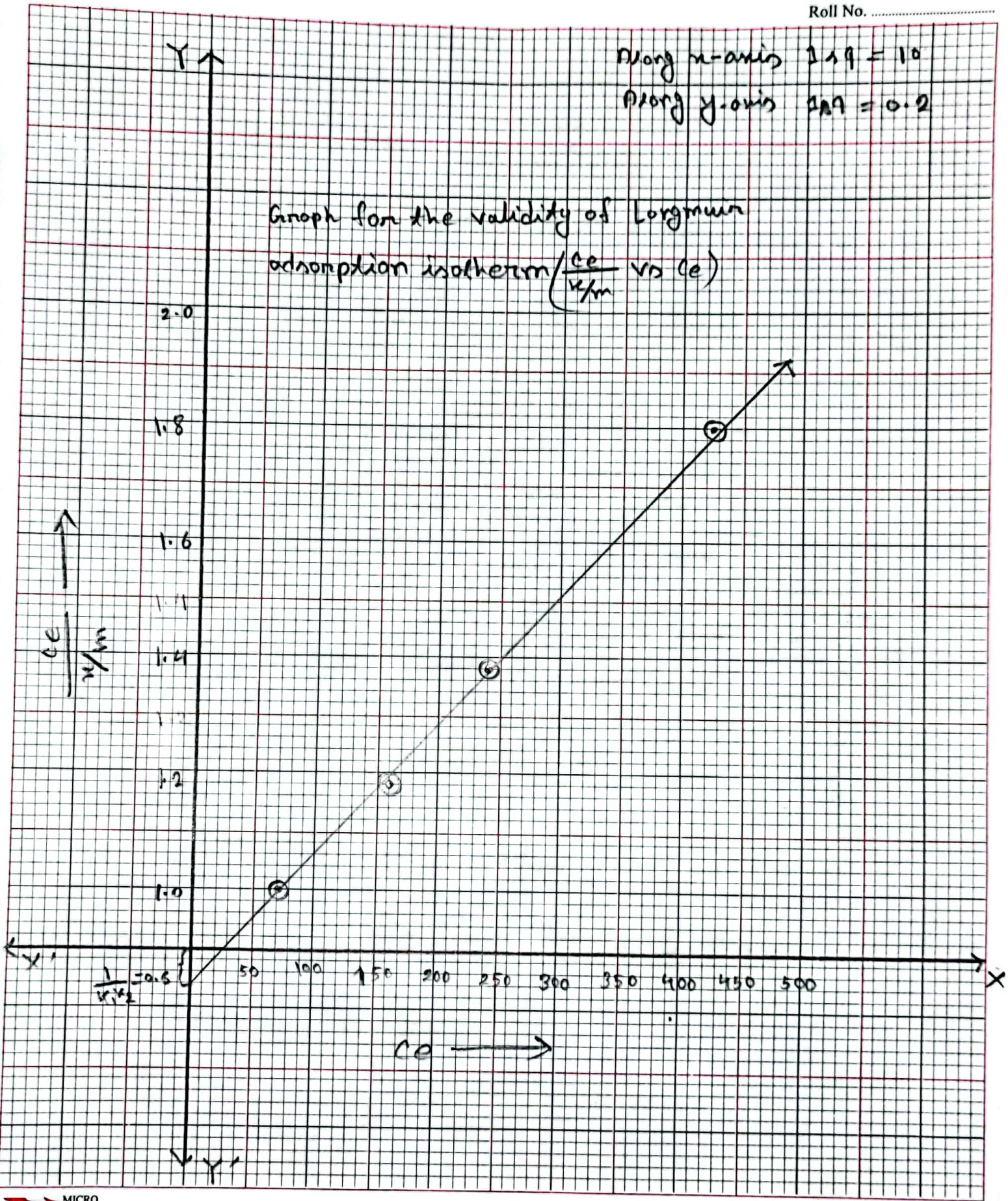
8 Inch x 10 Inch

Roll No. .....

along x-axis 1 sq = 10

along y-axis 1 sq = 0.2

Graph for the validity of Langmuir adsorption isotherm ($\frac{Ce}{x/m}$ vs Ce)



MICRO

8 Inch x 10 Inch

Data table for validity of freundlich and langmuir adsorption isotherm:

Bottle No	Ce milieq/L	x/m	log Ce	log x/m	Ce / x/m
1	422	234	2.62	3.36	1.80
2	320	240	2.50	2.38	1.33
3	245	165	2.39	2.21	1.48
4	193	171	2.28	2.23	1.12
5	156	132	2.19	2.12	1.18
6	75	75	1.8	1.8	1

Calculation from graph:

For Freundlich adsorption isotherm:

$$\log x/m = \log k + \frac{1}{n} \log c$$

From graph - 1:

$$\text{Slope} = \frac{PQ}{PR} = \frac{2.38 - 2.12}{2.50 - 2.19}$$

$$\Rightarrow \frac{1}{n} = 0.83$$

$$\therefore n = 1.20$$

The intercept, $\log K = 1.67$

Freundlich adsorption isotherm is valid.

For Langmuir adsorption isotherm,

$$\frac{C}{x/m} = \frac{1}{K_1 K_2} + \frac{C_e}{K_2}$$

From graph-2,

$$\text{slope} = \frac{1.88 - 1.18}{245 - 156} = 0.0079$$

$$\therefore \frac{1}{K_2} = 0.0079$$

$$\therefore K_2 = 126.58$$

And the intercept, $\frac{1}{K_1 K_2} = -0.6$

The Langmuir isotherm is valid.

Result:

1. The validity of Freundlich adsorption isotherm is studied by plotting of $\log x/m$ vs $\log ce$ in graph-1 which gives a straight line with slope $\frac{1}{n} = 0.83$ and intercept is, $\log K = 1.67$.

This is indicate validity of Freundlich adsorption isotherm.

2. The validity of Langmuir adsorption isotherm is studied by plotting of $\frac{ce}{x/m}$ vs ce in graph-2 which gives a straight line with slope, $\frac{1}{K_2} = 0.0079$ and the intercept $\frac{1}{K_1 K_2} = -0.6$.

This is indicates the validity of Langmuir adsorption isotherm.

Discussion:

1) Before starting the experiment, I cleaned all instrument properly.

2) Using acetic acid, phenolphthalein indicator, I found out the concentration of NaOH.

3) The weight of the charcoal carefully and the shaking process is done for 30 min.

4) At the begining of each experiment and after being adsorbed, I determined the concentration of acetic acid.

5) From this experiment we see that, the extent of absorption is directly proportional to the concentration of the solution.

6) For titration small filter paper is used to minimize the error due to the adsorption of acid by filter paper.

Date: 08-08-23
Group no- 08
Roll no-205

Experiment No-03

Name of the experiment-

Studying the liquid-liquid miscibility of three liquid system and determination of the position of one tie line.

Data:

Data table for the determination of the liquid liquid miscibility of the three liquid system and the position of tie line.

% of $CHCl_3$	weight of $CHCl_3$ (gm)	Volm of $CHCl_3$ $V=\frac{m}{D}$ / $V=\frac{m}{1.5}$ (mL)	weight of CH_3COOH (gm)	Volm of CH_3COOH $V=\frac{m}{1.05}$ (mL)	Volm of water (mL)
10	2	1.33	18	17.14	33
20	4	2.67	16	15.24	13
30	6	4.00	14	13.33	7.4
40	8	5.33	12	11.43	5.3
50	10	6.67	10	9.52	3.6
60	12	8.00	8	7.62	2.3
70	14	9.33	6	5.71	0.9
80	16	10.67	4	3.81	0.4
90	18	12.00	2	1.90	0.3

S. Hamadi 8.8.23

1. Room temperature-

Calculation:

Determination of the percentage of $CHCl_3$, CH_3COOH and H_2O in the mixture:

Bottle No	Total volume	% of $CHCl_3$	% of CH_3COOH	% of H_2O
10% $CHCl_3$ + 90% CH_3COOH	1.33 + 17.14 + 33 = 51.47	$\frac{1.33 \times 100}{51.47}$ = 2.58	$\frac{17.14 \times 100}{51.47}$ = 33.30	$\frac{33 \times 100}{51.47}$ = 64.11
20% $CHCl_3$ + 80% CH_3COOH	2.67 + 15.24 + 13 = 30.91	$\frac{2.67 \times 100}{30.91}$ = 8.63	$\frac{15.24 \times 100}{30.91}$ = 49.30	$\frac{13 \times 100}{30.91}$ = 42.05
30% $CHCl_3$ + 70% CH_3COOH	4 + 13.33 + 7.4 = 24.73	$\frac{4 \times 100}{24.73}$ = 16.13	$\frac{13.33 \times 100}{24.73}$ = 52.56	$\frac{7.4 \times 100}{24.73}$ = 29.92
40% $CHCl_3$ + 60% CH_3COOH	5.33 + 11.43 + 5.3 = 22.06	$\frac{5.33 \times 100}{22.06}$ = 24.16	$\frac{11.43 \times 100}{22.06}$ = 51.81	$\frac{5.3 \times 100}{22.06}$ = 24.02
50% $CHCl_3$ + 50% CH_3COOH	6.67 + 9.52 + 3.6 = 19.79	$\frac{6.67 \times 100}{19.79}$ = 33.84	$\frac{9.52 \times 100}{19.79}$ = 48.30	$\frac{3.6 \times 100}{19.79}$ = 18.36
60% $CHCl_3$ + 40% CH_3COOH	8 + 7.62 + 2.3 = 17.92	$\frac{8 \times 100}{17.92}$ = 44.64	$\frac{7.62 \times 100}{17.92}$ = 42.52	$\frac{2.3 \times 100}{17.92}$ = 12.83

Roll No. .....

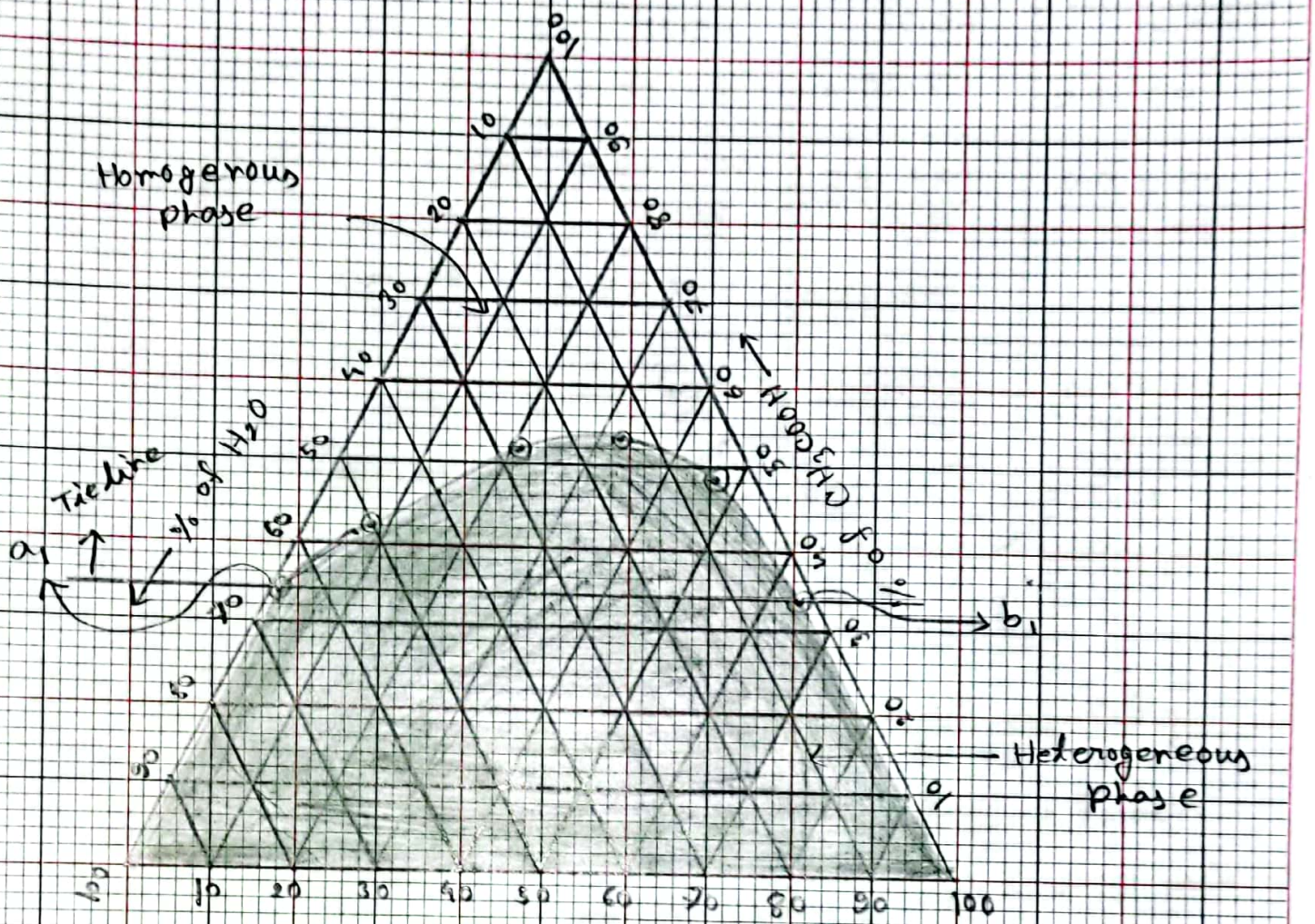


Fig: The ternary phase diagram of three component (H_2O, $CHCl_3$, CH_3COOH) system.

MICRO

8 Inch x 10 Inch

Bottle No	Total volume	% of $CHCl_3$	% of CH_3COOH	% of H_2O
70% $CHCl_3$ + 30% CH_3COOH	9.33 + 5.71 + 0.9 = 15.94	$\frac{9.33 \times 100}{15.94}$ = 58.53	$\frac{5.71 \times 100}{15.94}$ = 35.8	$\frac{0.9 \times 100}{15.94}$ = 5.64
80% $CHCl_3$ + 20% CH_3COOH	10.67 + 3.81 + 0.4 = 14.88	$\frac{10.67 \times 100}{14.88}$ = 77.70	$\frac{3.81 \times 100}{14.88}$ = 25.60	$\frac{0.4 \times 100}{14.88}$ = 2.68
90% $CHCl_3$ + 10% CH_3COOH	12 + 1.90 + 0.3 = 14.20	$\frac{12 \times 100}{14.20}$ = 84.50	$\frac{1.90 \times 100}{14.20}$ = 13.38	$\frac{0.3 \times 100}{14.20}$ = 2.11

Result :

1. Curve obtain from joining experimentally determined point from the boundary between homogeneous and heterogeneous mixture. A mixture of CH_3COOH inside the triangle will form homogeneous solution while any mixture represent in two layers it is required to draw the line across the miscibility indicate solution pair.

2. The position of line is represented in the diagram. The point corresponding to the following composition.

At point a_1,

% of $CHCl_3$ = 58.53

% of CH_3COOH = 35.80

% of H_2O = 5.64

At point b_1,

% of $CHCl_3$ = 2.58

% of CH_3COOH = 33.30

% of H_2O = 64.11

Discussion:

1. $CHCl_3$ is a volatile compound, so during preparation of the mixture of $CHCl_3$ and CH_3COOH, some amount of $CHCl_3$ may vaporised. So, we may not find the exact volume of $CHCl_3$.

2. In this experiment we have used glacial acetic acid, because glacial acetic acid do not contain any water. There all apparatus was moisture free.

3. Before starting the experiment, we dried all instrument properly.

4. we observed the turbidity of solution carefully.

5. Finally calculate the percent composition of mixture properly.

Group no- 08
Date : 14-08-23
Roll no - 205

Experiment NO - 04

Name of the experiment:

Determination of the partial molal volume of solution (Alcohol-water system) by westphal balance.

Data table:

Data table for the determination of weight lost at different concentration of alcohol solution -

% of alcohol solution	Description of weight	weight lost
0 %	(0.7×0.1) + (0.9×0.04) + (0.1×5) + (0.8×5) + (0.3×1)	4.900
15%	(0.1×5) + (0.8×5) + (0.3×1) + (0.6×0.05)	4.830
30%	(0.2×1) + (0.9×5)	4.700
45%	(0.2×0.5) + (0.7×15)	4.600
60%	(0.3×15)	4.500
75%	(0.2×15) + (0.4×1) + (0.5×0.5) + (0.6×0.5) + (0.8×0.5)	4.350
90%	(0.6×5) + (0.8×1)	3.800
100%	(0.3×0.2) + (0.7×5) + (0.4×0.1)	3.600

Calculation:

Table:
Calculation for the specific volume of alcohol and water mixture.

weight loss in water (gm)	% of alcohol in mixture	weight of alcohol + water	Density $\rho = \frac{W_{alc}}{W_{H_2O}}$	Specific volume, $\frac{1}{\rho}$ (mL/g)
4.90	0	4.90	1.00	1.00
4.90	15	4.83	0.98	1.02
4.90	30	4.70	0.96	1.04
4.90	45	4.60	0.94	1.06
4.90	60	4.50	0.92	1.09
4.90	75	4.35	0.89	1.12
4.90	90	3.80	0.77	1.29
4.90	100	3.60	0.73	1.37

From the specific volume vs percentage of alcohol plot, it is notice that partial specific volume of H_2O = 0.96, partial specific volume of methanol = 1.17.

Hence, partial molal volume of $H_2O = \frac{1}{\rho} \times m$

$= 0.96 \times 18 \text{ mL mol}^{-1}$

$= 17.28 \text{ mL mol}^{-1}$

∴ Partial molal volume of methanol,

$= \frac{1}{\rho} \times M = 1.17 \text{ mL} \cdot \text{g}^{-1} \times 32 \text{ g mol}^{-1}$

$= 37.44 \text{ mL} \cdot \text{mol}^{-1}$

Roll No. .....

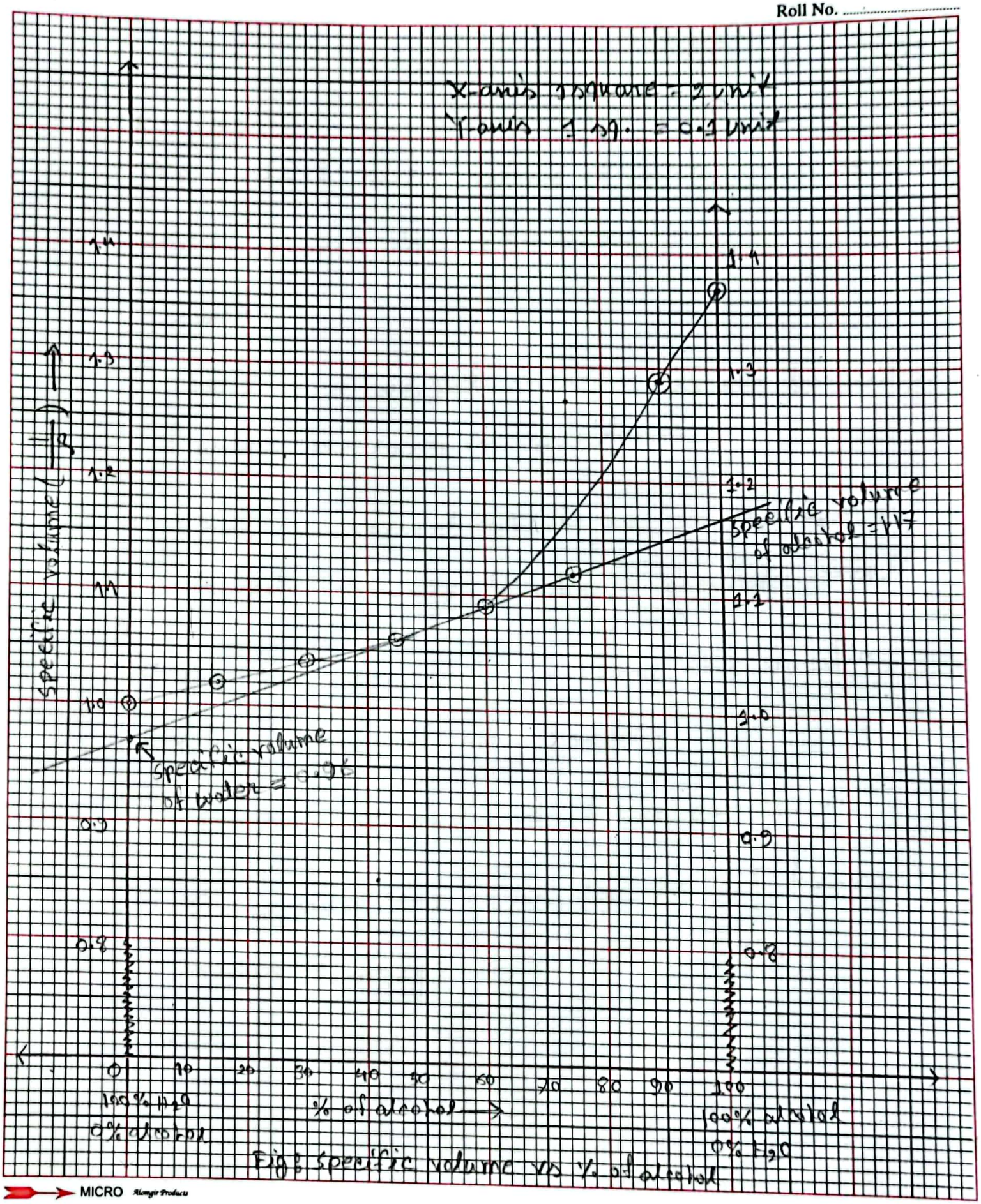


Fig: Specific volume vs % of alcohol

MICRO Alamgir Products

8 Inch x 10 Inch

Result:

The partial molal volume of water is 17.25 mL mol^{-1} and methanol is 37.44 mL mol^{-1}

Discussion:

① The loss of weight increases as the percentage of water increases in mixture.

② This experiment should be carried out air protected laboratory to obtain accurate result.

③ The loss of water and hence the specific volume does not vary linearly.

④ The measurement was taken carefully.

⑤ There may some error due to fail taking reading timely.

Date: 20-08-23
Group no-08
Roll no-205

Experiment No-05

Name of the experiment:

Determination of the partial molal volume of solution (Alcohol-water system) by pyenometer.

Data for this experiment:

1. Mass of dry pyenometer = 15.47 g
2. Mass of pyenometer with water = 25.57 g
3. Mass of pyenometer with 15% methanol = 25.34 g
4. Mass of pyenometer with 30% methanol = 25.17 g
5. Mass of pyenometer with 45% methanol = 24.87 g
6. Mass of pyenometer with 60% methanol = 24.57 g
7. Mass of pyenometer with 75% methanol = 24.23 g
8. Mass of pyenometer with 90% methanol = 23.81 g
9. Mass of pyenometer with 100% methanol = 23.49 g
10. Mass of water, $m_{H_2O} = (25.57 - 15.47)$ g
$= 10.10$ g

20/08/23

Calculation:

Table-1: Data table for the calculation of mole fraction of water (A) and alcohol (B):

Aqueous methanol Soln	Wt. of methanol W_B (g)	No. of mole (CH_3OH) $n_B = \frac{W_B}{M_B}$	wt of water W_A (g)	No. of mole (H_2O) $n_A = \frac{W_A}{M_A}$	Mole fraction of methanol $X_B = \frac{n_B}{n_A + n_B}$	Mole fraction of water $X_A = 1 - X_B$
water	0	0.00	100	5.56	0.00	1.00
15%	15	0.47	85	4.72	0.09	0.91
30%	30	0.94	70	3.89	0.19	0.81
45%	45	1.41	55	3.06	0.32	0.68
60%	60	1.88	40	2.22	0.46	0.54
75%	75	2.34	25	1.39	0.63	0.37
90%	90	2.81	10	0.56	0.83	0.17
100%	100	3.13	0	0.00	1.00	0.00

Table-2:

Data table for the determination of the molar volume of alcohol (B) and water (A):

Solution (Aq)	For CH_3OH $x_B = 1 - x_A$	For H_2O x_A	For solution $M_{av} = x_A M_A + (1-x_A) M_B$ g mol^{-1}	Mass of liquid + Pyrometer (g)	Mass of liquid m_2 (g)	For Sol^n density $d_2 = \frac{m_2}{m_{H_2O}} d_{H_2O}$ g cm^{-3}	Molar volume $V_m = \frac{M_{av}}{d_2}$ cm^3/mol
water	0.0	1.00	18.00	25.57	10.10	1.0	18.00
15% MeOH	0.09	0.91	19.26	25.34	9.87	0.98	19.65
30% MeOH	0.19	0.81	20.66	25.17	9.70	0.96	21.52
45% MeOH	0.32	0.68	22.48	24.87	9.40	0.93	24.17
60% MeOH	0.46	0.54	24.44	24.57	9.10	0.90	27.15
75% MeOH	0.63	0.37	26.82	24.23	8.76	0.87	30.83
90% MeOH	0.83	0.17	29.62	23.81	8.34	0.82	36.12
100% MeOH	1.00	0.00	32.00	23.49	8.02	0.79	40.5

From the graph,

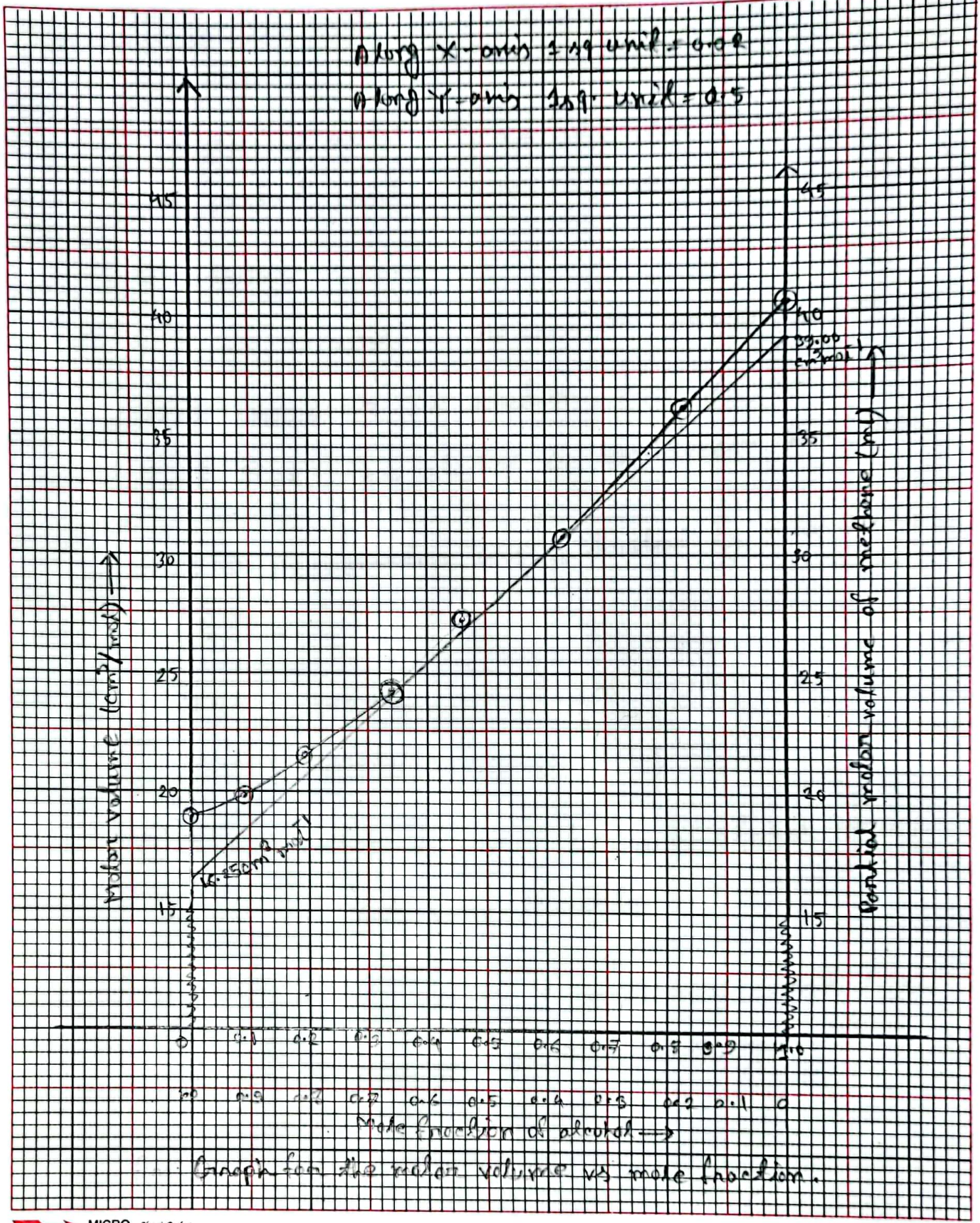
The partial molal volume of water is $= 16.25 cm^3 mol^{-1}$

The partial molal volume of methanol $= 39.00 cm^3 mol^{-1}$

Roll No. .....

along x-axis 1 big unit = 0.02

along y-axis 1st. unit = 0.5



Graph for the molar volume vs mole fraction.

MICRO Alamgir Products

8 Inch x 10 Inch

Result:

The partial molal volume of water is $16.25 cm^3 mol^{-1}$

The partial molal volume of methanol is $39.00 cm^3 mol^{-1}$

Discussion:

① Molal volume increases as percentage of methanol increases in solution, this is due to the fact that, the alcohol is less denser than water.

② The pycnometer was calibrated by measuring solution.

③ The pycnometer was filled accurately on weight carefully.

④ The specific volume solution of alcohol and water does not vary linearly.

⑤ The partial molar volume was found from graph which was plotted carefully.

Date: 21-08-23
Group no-08
Roll no-205

Experiment NO-06

Name of the experiment:

Finding of the:

(a) specific rotation of sucrose

(b) Concentration of an unknown sucrose solution

(c) Influence of solvent on the rotation of sucrose solution by polarimetaric method.

Data:

Data for the measurement of rotation θ, of the sucrose solution by polarimeter.

Concentration of sucrose solution (g/100ml)	Circular scale rotation (degree)	Vernier scale reading (cm)	Vernier constant v.c	Total scale reading (degree)
5	9	2	0.05	9.1
10	18	0	0.05	18
15	29	0	0.05	29
20	38	5	0.05	38.25

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(b) For unknown solution found, θ = 20.12

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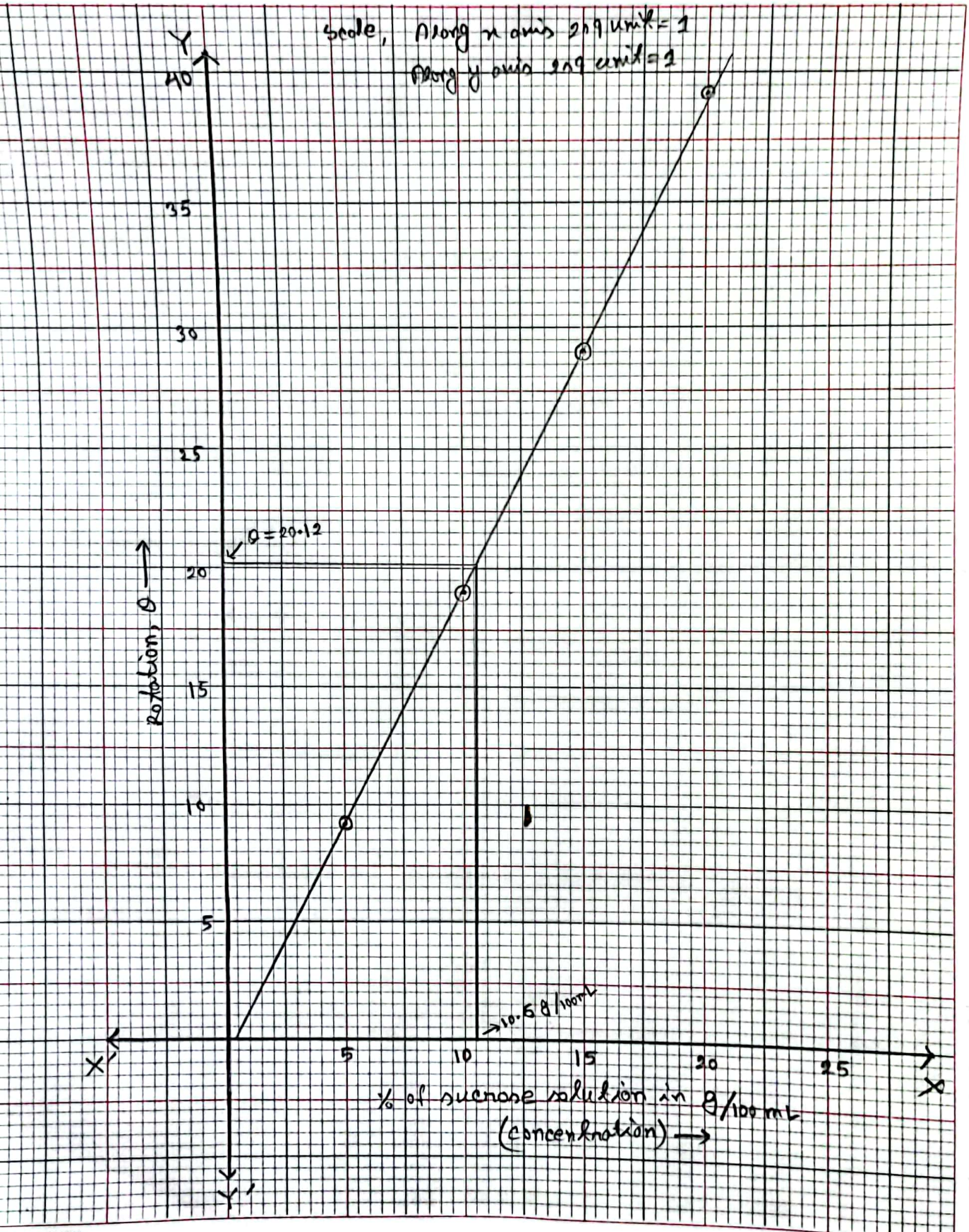

MICRO

8 Inch x 10 Inch

Calculations:

Calculations for determination of vernier constant of polarimeter:

Here,

Smallest reading on main scale = 1 cm

No. of division on vernier scale = 20

Vernier constant,

$$= \frac{\text{Smallest reading on main scale}}{\text{No of division on vernier scale}}$$

$$= \frac{1\,cm}{20}$$

$$= 0.05\,cm$$

(a) Specific rotation of sucrose solution is given by the equation:

$$[\alpha]^{t}_{\lambda} = \frac{100 \times \theta}{c \times l}$$

where,

l = length of polarimeter tube = 2dm

c = concentration of the sucrose solution

calculation for 5% solution:

In 5% solution, $c = 5\ g/100mL$

$l = 2\ dm$

$\theta = 9.1$ degree

$\therefore$ specific rotation, $[\alpha]_\lambda^t = \frac{100 \times \theta}{c \times l}$

$\Rightarrow [\alpha]_\lambda^t = \frac{100 \times 9.1}{5 \times 2}$

$= 91$ degree $mL\ g^{-1}\ dm^{-1}$

calculation for 10% solution:

10% solution means, $c = 10\ g/100mL$

$\theta = 18$ degre

$l = 2\ dm$

$\therefore$ specific rotation,

$[\alpha]_\lambda^t = \frac{100 \times \theta}{c \times l}$

$= \frac{100 \times 18}{10 \times 2}$

$= 90$ degree $mL\ g^{-1} dm^{-1}$

Calculation for 15% solution:

15% solution means,

$c = 15\ g/100mL$

$\theta = 29$ degree

$l = 2\ dm$

$\therefore$ specific rotation, $[\alpha]_\lambda^t = \frac{100 \times \theta}{c \times l}$

$= \frac{100 \times 29}{15 \times 2}$

$= 96.67\ degree\ mL\ g^{-1}\ dm^{-1}$

Calculation for 20% solution:

20% solution means,

$c = 20\ g/100mL$

$\theta = 38.25$ degree

$l = 2dm$

$\therefore$ specific rotation, $[\alpha]_\lambda^t = \frac{100 \times \theta}{c \times l}$

$= \frac{100 \times 38.25}{20 \times 2}$

$= 95.62\ degree\ mL\ g^{-1}\ dm^{-1}$

The average specific rotation of sucrose solution

$$\text{is} = \frac{91 + 90 + 96.67 + 95.82}{4}$$

$$= 93.32 \text{ degree mL g}^{-1} \text{dm}^{-1}$$

The actual specific rotation of sucrose solution

$$\text{is} = 66.6 \text{ degree mL g}^{-1} \text{dm}^{-1}$$

$$\% \text{ of error} = \frac{\text{Actual value} - \text{Experimental value}}{\text{Actual value}} \times 100\%$$

$$= \frac{93.32 - 66.6}{66.6} \times 100\%$$

$$= 40.12\%$$

(b) For unknown solution found $\theta = 20.12$ degree.

From graph the concentration of unknown solution is 10.6 g/100mL

Result:

(a) The specific rotation of 5%, 10%, 15%, 20% solution are 91 degree $mL\,g^{-1}\,dm^{-1}$, 90 degree $mL\,g^{-1}\,dm^{-1}$, 96.67 degree $mL\,g^{-1}\,dm^{-1}$, 95.62 degree $mL\,g^{-1}\,dm^{-1}$ respectively.

The average specific rotation is 93.32 degree $mL\,g^{-1}\,dm^{-1}$.

% of error 40.12 %.

(b) concentration of the unknown solution is 10.6 g/100mL

(c) The influence of solvent is the lowering of the rotation of the sucrose solution.

Discussion:

(1) From this experiment and calculating data we find the specific rotation of the sucrose soln is +93.32 that means the sucrose solution optically active and dextrorotatory.

(2) I cleaned the polarimeter tube carefully.

(3) In this experiment the used water was dust free.

(4) Whenever a solution is changed, I rinsed the tube with the new solution under examination.

(5) There was no air bubble inside the tube.

(6) The position of analyzer was set accurately.

(7) The temperature and wavelength was fixed.

(8) The sucrose solution were prepared carefully.

Date: 22-08-23
Group no-08
Roll no-205

Experiment No-07

Name of the experiment:

Determination of the (a) absorption curve of coloured substance in solution, (b) the validity of Beer-Lambert law, and (c) the strength of an unknown solution.

Data:

Data for the determination of λ_{max} value for 10% given $KMnO_4$ coloured substance.

Wavelength (nm)	Absorbance	Wabelength (nm)	Absorbance
480	188	540	468
490	260	550	456
500	344	560	310
510	406	570	275
520	481	580	152
530	500	590	78

Data for the determination of the absorbance of different concentration of given coloured substance at $\lambda_{max} = 530$

λ_{max} (nm)	conc. of $KMnO_4$ soln (M)	Absorbance
530	5×10^{-5}	147
530	10×10^{-5}	258
530	15×10^{-5}	375
530	20×10^{-5}	500
530	25×10^{-5}	621

29.8.23

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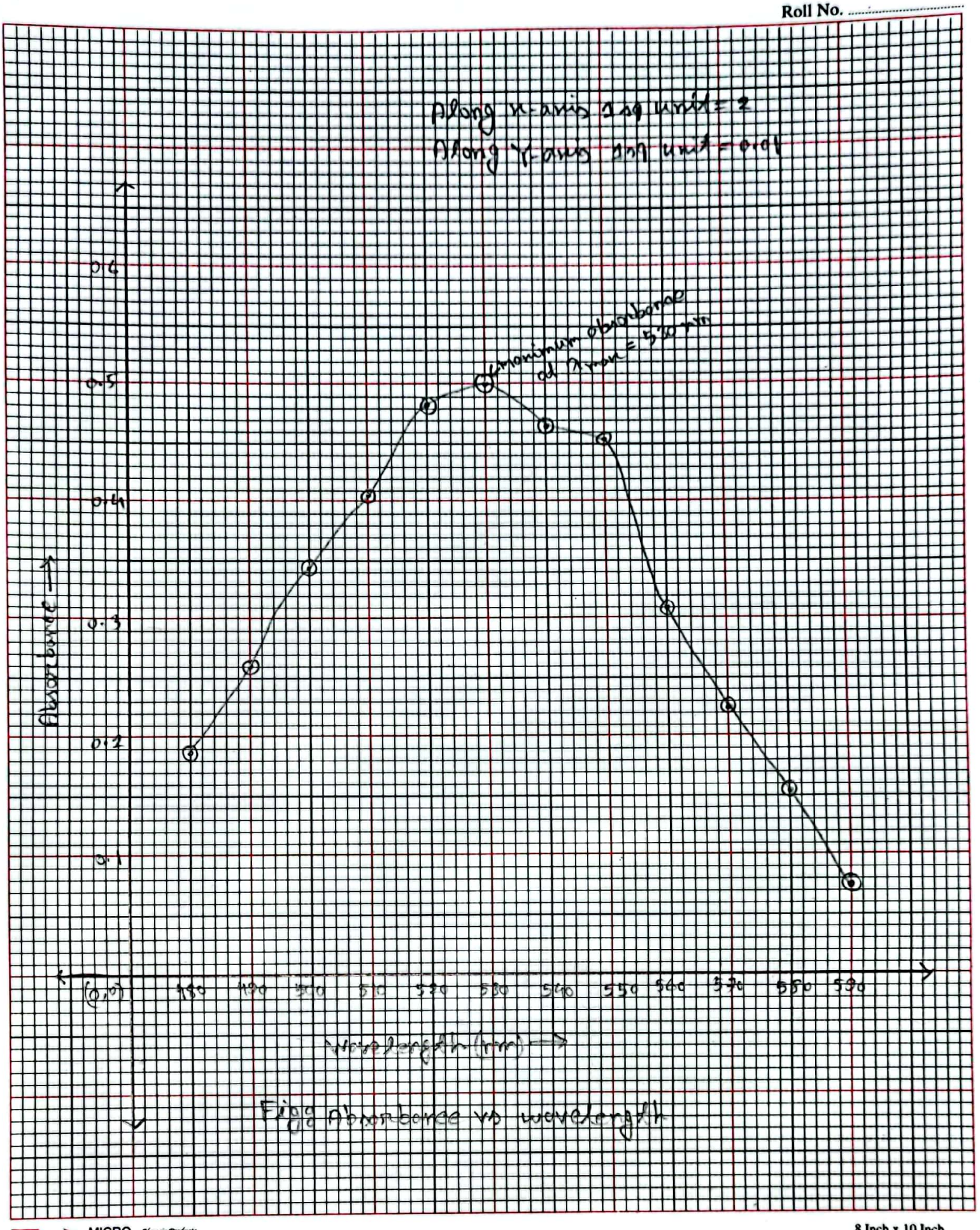
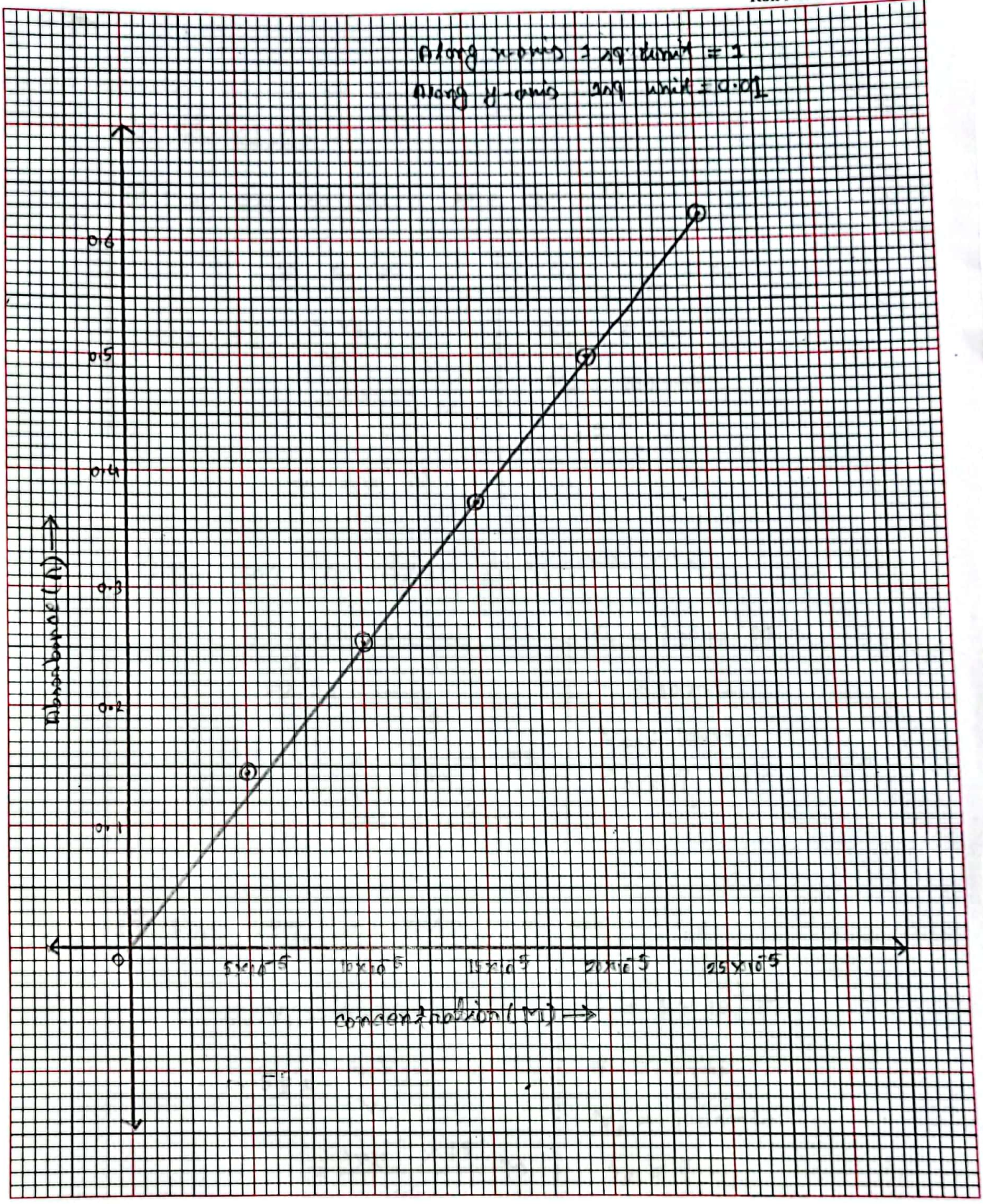


Fig: Absorbance vs wavelength

Roll No. .....


0.6
0.5
0.4
0.3
0.2
0.1
0
5x10⁻⁵
10x10⁻⁵
15x10⁻⁵
20x10⁻⁵
25x10⁻⁵
concentration (M) →
absorbance (A) →

Calculations:

① For 5% solution the conc. is,

$$S_1V_1 = S_2V_2$$

$$\Rightarrow S_2 = \frac{S_1V_1}{V_1}$$

$$= \frac{0.001 \times 5}{100} \text{ M}$$

$$= 5 \times 10^{-5} \text{ M}$$

$S_1 = \frac{1}{1000}$ M

$V_1 = 5$ mL

$V_2 = 100$ mL

$S_2 = ?$

② For 10% solution the conc. is,

$$S_1V_1 = S_2V_2$$

$$\Rightarrow S_2 = \frac{S_1V_1}{V_2}$$

$$= \frac{1 \times 10^{-3} \times 10}{100} \text{ M}$$

$$= 10 \times 10^{-5} \text{ M}$$

$S_1 = 1 \times 10^{-3}$ M

$V_1 = 10$ mL

$V_2 = 100$ mL

$S_2 = ?$

③ For 15% solution the conc. is,

$$S_1V_1 = S_2V_2$$

$$\Rightarrow S_2 = \frac{S_1V_1}{V_2}$$

$$= \frac{1 \times 10^{-3} \times 15}{100} \text{ M}$$

$$= 15 \times 10^{-5} \text{ M}$$

$S_1 = 1 \times 10^{-3}$ M

$V_1 = 15$ mL

$V_2 = 100$ mL

$S_2 = ?$

④ For 20% solution the conc. is,

we know,

$$S_2 = \frac{S_1 V_1}{V_2}$$

$$= \frac{0.001 \times 20}{100} M$$

$$= 20 \times 10^{-5} M$$

$S_1 = 0.001 M$
$V_1 = 20 mL$
$V_2 = 100 mL$
$S_2 = ?$

⑤ For 25% solution the conc. is,

$$S_2 = \frac{S_1 V_1}{V_2}$$

$$= \frac{1 \times 10^{-3} \times 25}{100} M$$

$$= 25 \times 10^{-5} M$$

$S_1 = 1 \times 10^{-3} M$
$V_1 = 25 mL$
$V_2 = 100 mL$
$S_2 = ?$

i) From the graph, absorption maximum $\lambda_{max} = 530nm$.

ii) Beer-Lambert law is verified because the straight line pass through the origin.

iii) The absorbance found for the unknown solution is , so the concentration we get from the graph is

So, the concentration of the unknown solution is

Result:

i) From graph of absorption curve $\lambda_{max} = 530\,nm$

ii) Graph-2 show the validity curve of Beer-Lambert law. Because the straight line passes through the origine. [According to, $A = \epsilon c l$]

iii) The concentration of unknown solution is

Discussion:

(1) All the solution was prepared carefully.

(2) while determining λ_{max} for $KMnO_4$ water was used as a blank.

(3) λ_{max} was determine carefully.

(4) There were not present any air bubble in the tube containing $KMnO_4$ solution.

(5) From the absorption vs concentration graph we see that the extend of absorption is directly proportional to the amount of $KMnO_4$ in the solution present.

(6) There may have some error in this experiment

Date: 24-07-23
Group no-08
Roll no-205

Experiment NO-08

Name of the experiment:

Determination of the specific reaction rate of base catalysed hydrolysis of an ester by conductometric method.

Data table:

Data for the determination of the specific reaction rate of a base catalysed hydrolysis of an ester by conductometric method:

Time (min)	Conductance ($mS\,cm^{-1}$)	Time (min)	Conductance (mS)	Time (min)	Conductance (mS)
0.5	7.30	7.0	4.26	13.5	3.80
1.0	6.75	7.5	4.20	14.0	3.79
1.5	6.19	8.0	4.15	14.5	3.77
2.0	5.77	8.5	4.10	15.0	3.76
2.5	5.46	9.0	4.06	15.5	3.74
3.0	5.21	9.5	4.02	16.0	3.73
3.5	5.01	10.0	3.98	16.5	3.72
4.0	4.85	10.5	3.95	17.0	3.71
4.5	4.71	11.0	3.92	17.5	3.70
5.0	4.60	11.5	3.89	18.0	3.69
5.5	4.49	12.0	3.86	18.5	3.68
6.0	4.41	12.5	3.84	19.0	3.67
6.5	4.33	13.0	3.82	19.5	3.66

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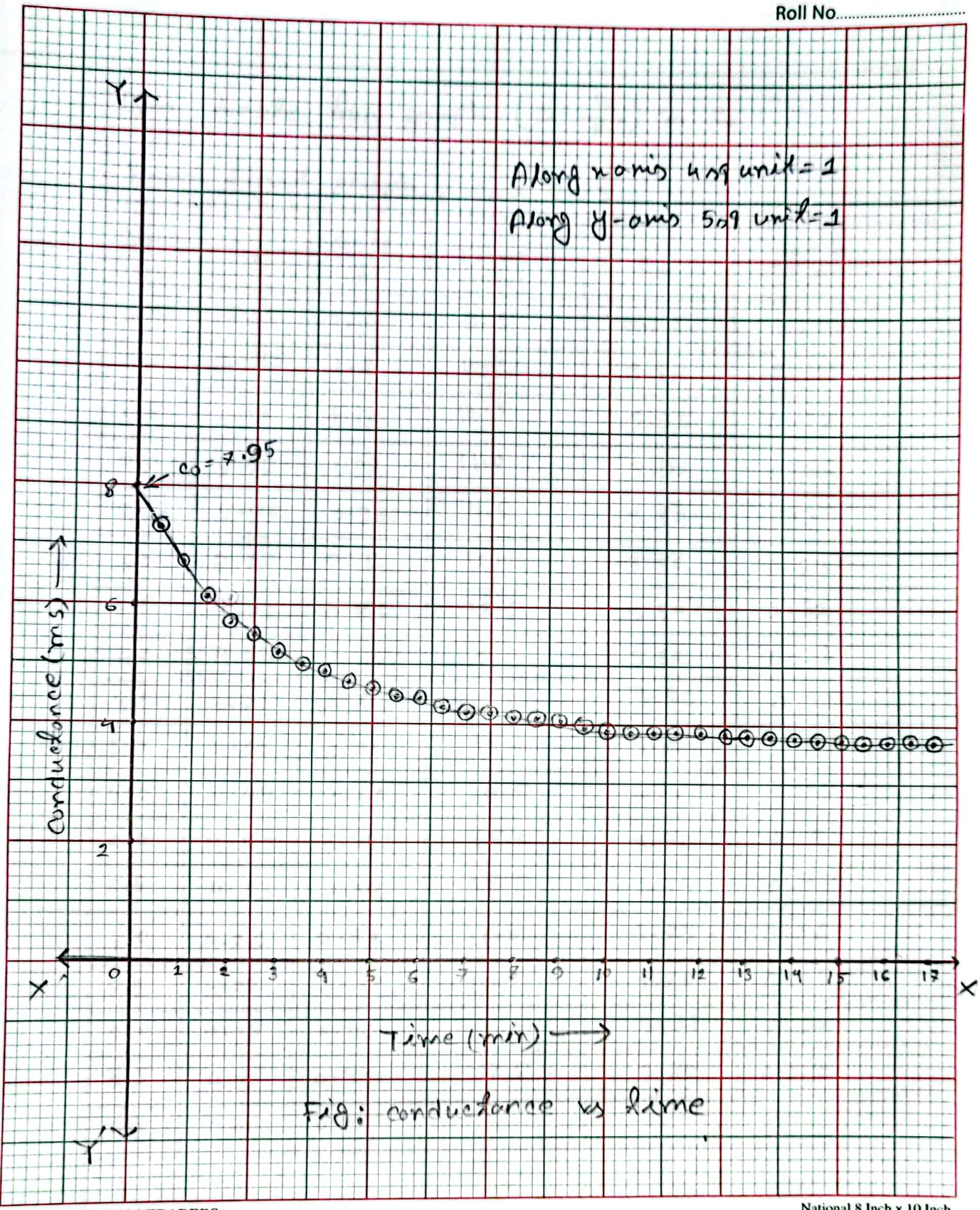


Fig: conductance vs time.

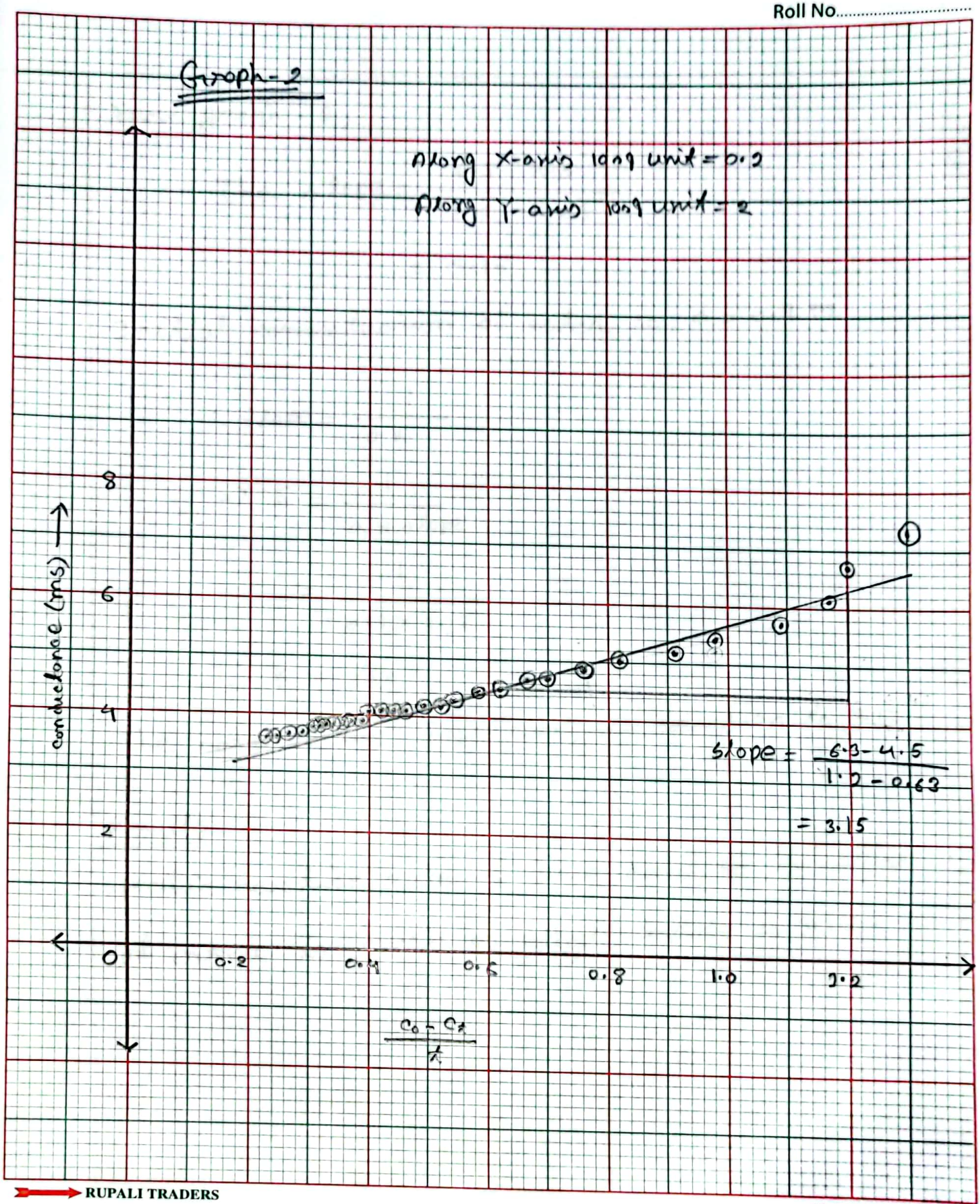
Data table for graph $(c_0 - \frac{c_t}{t})$ vs c_t.

From first graph, $c_0 = 7.95$

Time (min)	c_t	$\frac{c_0 - c_t}{t}$	Time (min)	c_t	$\frac{c_0 - c_t}{t}$
0.5	2.30	1.3	9.5	4.02	0.40
1.0	6.75	1.2	10.0	3.98	0.39
1.5	6.19	1.17	10.5	3.95	0.37
2.0	5.77	1.09	11.0	3.92	0.36
2.5	5.46	0.996	11.5	3.89	0.34
3.0	5.21	0.913	12.0	3.86	0.33
3.5	5.01	0.82	12.5	3.84	0.32
4.0	4.85	0.76	13.0	3.82	0.31
4.5	4.71	0.708	13.5	3.80	0.30
5.0	4.60	0.66	14.0	3.79	0.29
5.5	4.49	0.62	14.5	3.77	0.28
6.0	4.41	0.581	15.0	3.76	0.27
6.5	4.33	0.54	15.5	3.74	0.27
7.0	4.26	0.52	16.0	3.73	0.26
7.5	4.20	0.49	16.5	3.72	0.25
8.0	4.15	0.46	17.0	3.71	0.24
8.5	4.10	0.44	17.5	3.70	0.23
9.0	4.06	0.42			

Roll No.....

Graph-2



From graph - 2,

$$\text{The slope} = \frac{6.3 - 4.5}{1.2 - 0.63}$$

$$= 3.15$$

Initial concentration of ester is, $a = 0.1\ M$

$$\therefore \text{slope} = \frac{1}{ak}$$

$$\Rightarrow k = \frac{1}{a \times \text{slope}}$$

$$= \frac{1}{0.1 \times 3.15}$$

$$\therefore k = 3.17$$

The rate of the base catalysed reaction is 3.17

Result:

Specific reaction rate of base catalysed hydrolysis of ester is much as ethyl acetate is 3.17.

Discussion:

① At first we cleaned all the apparatus.

② The initial concentration and initial conductance of the mixture, was measured carefully from the graph.

③ All the reading were taken considering as such as care.

④ The conductance of the reaction mixture decreased with increasing of time.

⑤ There may be some instrumental error.

Date: 25-07-23
Group NO-08
Roll NO-205

Experiment NO-09

Name of the experiment:

Finding the equivalent conductance at infinite dilution and the dissociation constant of a weak electrolyte.

Experimental data:

Table for the conductance at infinite dilution of electrolyte:

No. of observation	conc. of CH_3COOH (N)	conductance ($\mu s/cm$)
1	0.00 (distilled H_2O)	2.6
2	0.01	310
3	0.05	270
4	0.033	200
5	0.005	70
6	0.0025	51
7	0.001	33
8	0.0005	26
9	0.0001	9

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29.6.23

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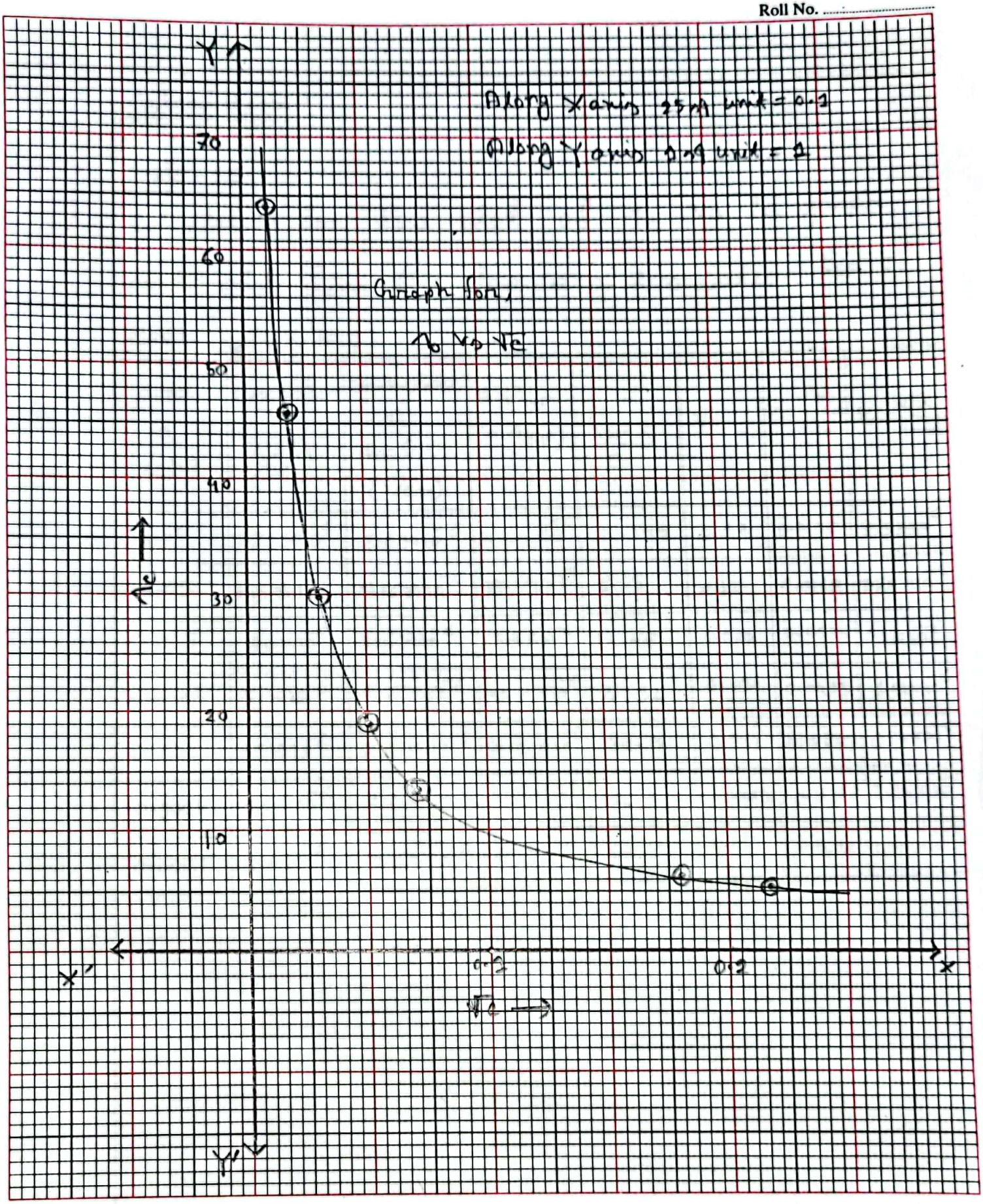

Along X axis 25 mm unit = 0.1
Along Y axis 5 mm unit = 2
Graph for
λ_0 Vs Ve
Y
70
60
50
40
30
20
10
Y'
Ve
X'
X
0.2
$\sqrt{\lambda_0}$

Calculation:

Table-1:

Calculation for the plot of Λ_c vs $\sqrt{c}$ graph and also find the degree of dissociation ~~for~~ of each solution:

conc. c (N)	Conductance (S/cm)	conductance of H_2O (S/cm)	$K = (I_{soln.} - I_{H_2O})$	$\Lambda_c = \frac{1000K}{c}$	$\sqrt{c}$	$\alpha = \frac{\Lambda_c}{\Lambda_0}$	$K_c = \frac{\alpha^2 c}{1-\alpha}$
0.05	270×10^{-6}	2.6×10^{-6}	267.4×10^{-6}	5.348	0.22	0.0136	9.3×10^{-6}
0.03	200×10^{-6}	2.6×10^{-6}	197.4×10^{-6}	6.58	0.17	0.0168	8.72×10^{-6}
0.005	70×10^{-6}	2.6×10^{-6}	67.4×10^{-6}	13.48	0.070	0.0344	6.1×10^{-6}
0.0025	51×10^{-6}	2.6×10^{-6}	48.4×10^{-6}	19.36	0.050	0.0495	6.49×10^{-6}
0.001	33×10^{-6}	2.6×10^{-6}	30.4×10^{-6}	30.4	0.031	0.0777	6.54×10^{-6}
0.0005	26×10^{-6}	2.6×10^{-6}	23.4×10^{-6}	46.8	0.0223	0.1197	8.05×10^{-6}
0.0001	9×10^{-6}	2.6×10^{-6}	6.4×10^{-6}	64.0	0.01	0.1637	3.20×10^{-6}

Table-2:

Table for the graph $\Lambda_e c$ vs $\frac{1}{\Lambda_e}$ and finding the equivalent conductance at infinite dilution.

conc. c (N)	Λ_e	$\Lambda_e \cdot c$	$\frac{1}{\Lambda_e}$
0.05	5.348	0.2674	0.1869
0.03	6.58	0.1974	0.1519
0.005	13.48	0.0674	0.074
0.00025	19.36	0.0484	0.0516
0.001	30.4	0.0304	0.032
0.0005	46.8	0.0234	0.021
0.0001	64.0	0.0064	0.0156

From graph,

$$\Lambda_e \cdot c \text{ vs } \frac{1}{\Lambda_e}$$

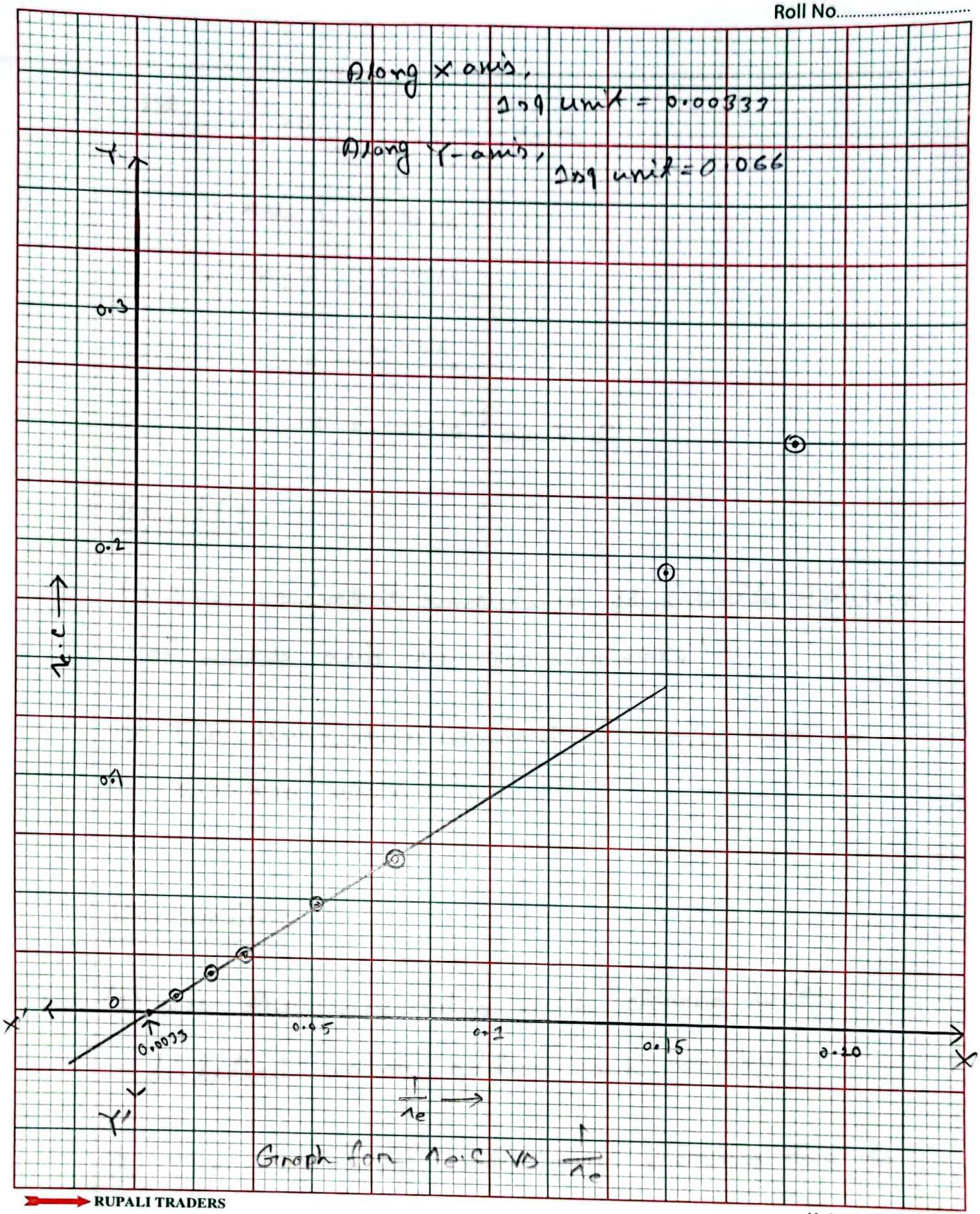
At $\Lambda_e \cdot c = 0$,

$$OX = \frac{1}{\Lambda_0} = 0.00333$$

$$\Rightarrow \Lambda_0 = 300 \text{ sem}^2 \text{equiv}^{-1}$$

and $OY = -0.0066$.

Roll No.....



Graph for $\lambda_e \cdot C$ Vs $\frac{1}{\lambda_e}$

RUPALI TRADERS

National 8 Inch x 10 Inch

At, $\frac{1}{\Lambda_e} = 0$ $\quad OY = -K_c \cdot \Lambda_0$

$\therefore$ $0 - K_c \Lambda_0 = -0.0066$

$\therefore$ $K_c = \frac{-0.0066}{\Lambda_0}$

$= \frac{-0.0066}{300}$

$= 2.20 \times 10^{-5}$ equiv. L^{-1}

Result:

i) The equivalent conductance at infinite dilution of acetic acid is 300 s.cm^{2}equiv^{-1}

ii) The dissociation constant of acetic acid is 2.20×10^{-5} equiv. L^{-1}.

Discussion:

(1) All the apparatus were cleaned with distilled water after each reading.

(2) The equivalent conductance increases due to mobile proton.

(3) All the solutions were prepared accurately and taken the part carefully.

(4) Conductance of distilled water used in the experiment and substrated it ~~fa~~ from the observed conductance of each solution to get specific conductance.

(5) Due to lacking of graph size therefore may have some error.

(6) At dilution increases the equivalent conductance is increases due to the degree of the dissociation -n remain the same, no more ion is produced in the system.

Date: 30.07.23

Group no-08

Roll no-205

Experiment NO-10:

Name of the experiment:

Conductometric titration of (a) strong and (b) weak acids and drawing their neutralisation curves.

Data table:

Data table for the standardization of NaOH against standard 0.1N succinic acid:

No of obs	Volm of succinic acid (mL)	Volume of NaOH Soln			Mean (mL)
		IBR(mL)	FBR (mL)	Diff (mL)	
1	10.00	5.0	10.0	5.0	5.0
2	10.00	10.0	15.0	5.0	
3	10.00	15.0	20.0	5.0	

Table-1:

conductometric titration of strong acid HCl with NaOH:

volume of NaOH (mL)	conductance (ms/cm)
0	27.00
1	22.08
2	17.56
3	13.30
4	12.10
5	16.20
6	20.46
7	24.47
8	28.44
9	32.27
10	36.24

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31.7.23

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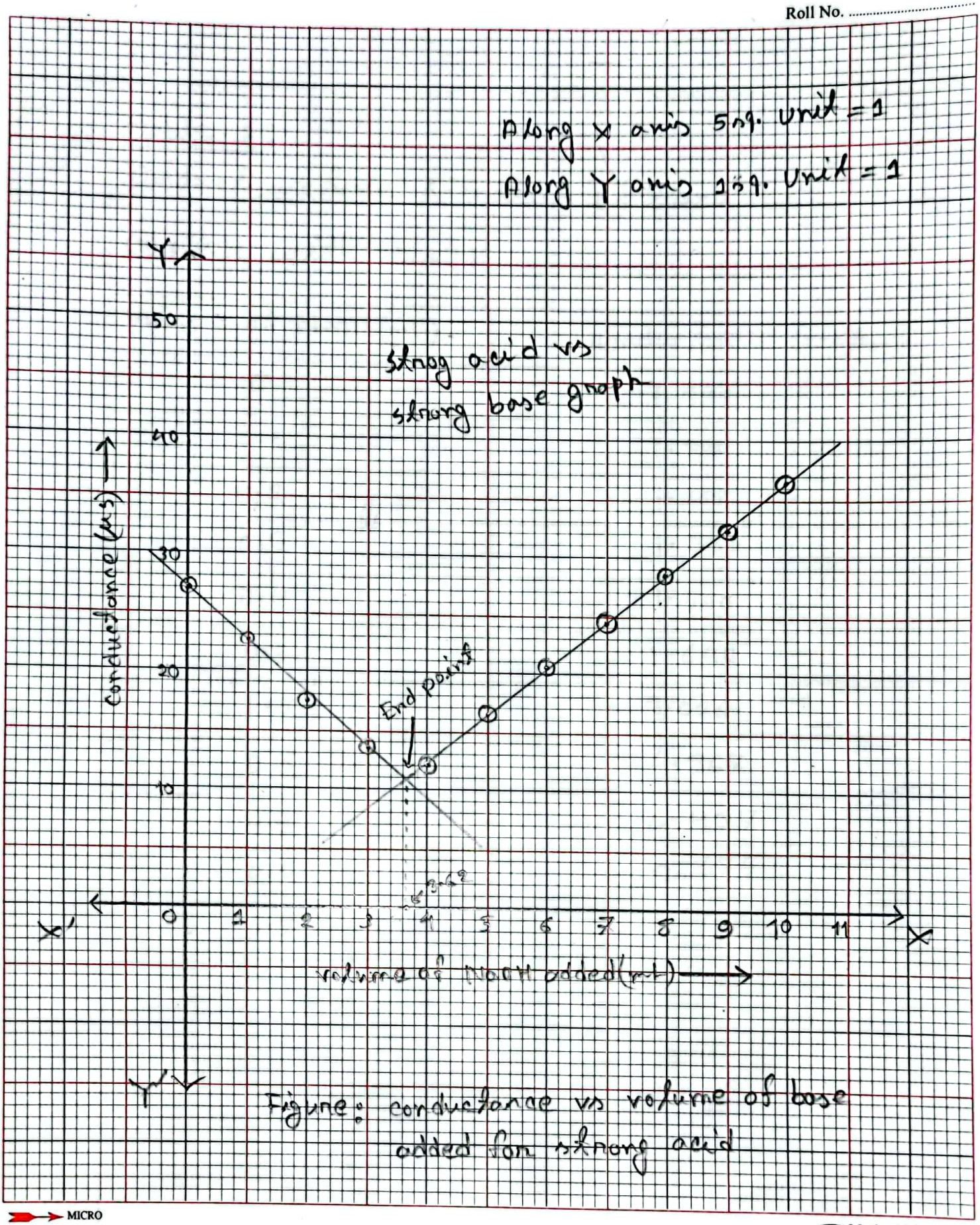

Figure: conductance vs volume of base added for strong acid

Table - 2:

Conductometric titration of weak acid CH_3COOH with NaOH

Volume of NaOH (mL)	conductance (ms/cm)
0	1.391
0.1	1.300
0.2	1.205
0.3	1.350
0.4	1.500
0.5	1.640
1.5	3.026
2.5	4.367
3.5	6.437
4.5	9.675
5.5	11.850

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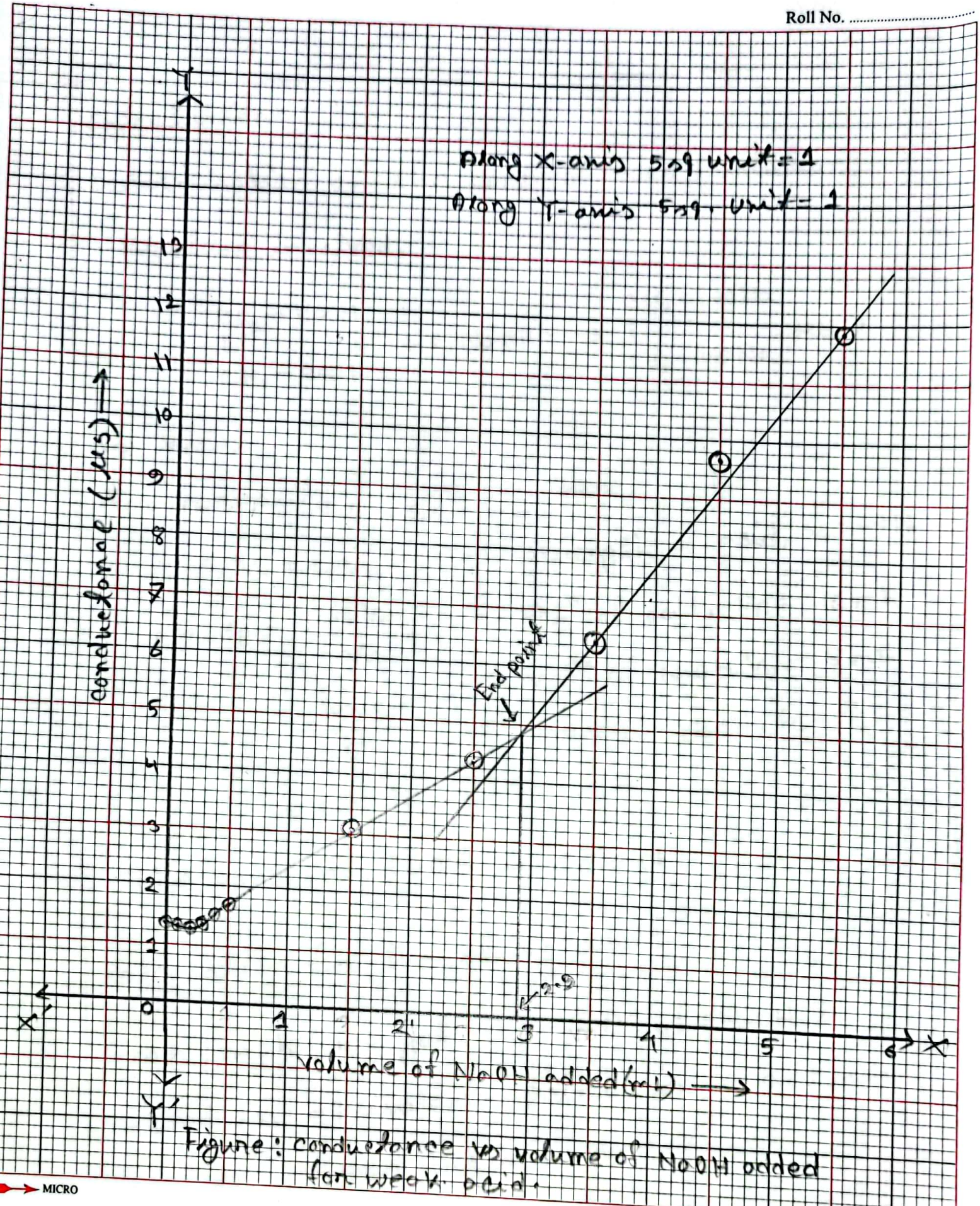

Figure: conductance vs volume of NaOH added for weak acid.

MICRO

NaCl + H

8 Inch x 10 Inch

Calculations:

Determination of the conc. of NaOH solution,

Reaction involved,

$$\begin{array}{l} CH_2-COOH \\ | \\ CH_2-COOH \end{array} + 2\,NaOH \longrightarrow \begin{array}{l} CH_2-COONa \\ | \\ CH_2-COONa \end{array} + 2H_2O$$

we know,

For normality,

$$S_A V_A = S_B V_B$$

$$\Rightarrow S_B = \frac{S_A V_A}{V_B}$$

$$= \frac{10 \times 0.1}{5}$$

$$= 0.2\ N$$

Here,

Strength of acid, $S_A = 0.1\ N$

Volᵐ of acid, $V_A = 10\ mL$

Volᵐ of NaOH $V_B = 5\ mL$

Conc. of NaOH, $S_B = ?$.

The conc. of NaOH is 0.2 N.

calculation for the strong acid:

From graph, we get the volᵐ of NaOH is needed in equivalent point is ≈ 3.62 mL.

Reaction between HCl and NaOH is,

$$NaCl + HCl \longrightarrow NaCl + H_2O$$

We know,

$$S_A V_A = S_B V_B$$

$$\Rightarrow S_A = \frac{S_B V_B}{V_A}$$

$$= \frac{0.2 \times 3.62}{10}$$

$$= 0.0724 \text{ N}$$

Here,

volume of NaOH, $V_B = 3.62 \text{ mL}$

conc. of NaOH, $S_B = 0.2 \text{ N}$

volume of acid, $V_A = 10 \text{ mL}$

conc. of acid, $S_A = ?$

The conc. of strong acid (HCl) is $= 0.0724 \text{ N}$

Calculation for weak acid:

Reaction involved,

$$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$$

We know,

$$S_A V_A = S_B V_B$$

$$\Rightarrow S_A = \frac{S_B V_B}{V_A}$$

$$= \frac{0.2 \times 2.9}{10}$$

$$= 0.058 \text{ N}$$

Here,

From graph, the volume of the NaOH sol^n is, $V_B = 2.9 \text{ mL}$

Concentration of NaOH, $S_B = 0.2 \text{ N}$

vol^m of CH_3COOH, $V_A = 10 \text{ mL}$

Conc. of CH_3COOH, $S_A = ?$

The concentration of weak acid (CH_3COOH) is

$= 0.058 \text{ N}$

Result:

(a) The strength of NaOH is 0.2 N

(b) The strength of strong acid, HCl = 0.0724 N

(c) The strength of weak acid, CH_3COOH = 0.058 N

Discussion:

① Apparatus was cleaned by distilled water.

② Measurement was taken carefully and plotted the graph accurately.

③ NaOH is secondary standard substance, so at first it was standard with 0.1 N succinic acid.

④ When NaOH is added, the conductivity of HCl decreases due to the neutralization of highly mobile H^+ ions by OH^- ions of NaOH untill an equivalent point was reached.

⑤ On addition of base to CH_3COOH, At first conductivity was decreased due to common ion acetate and very soon the conductivity was increased due to the formation of strong electrolyte CH_3COONa.

Date: 31.07.23
Group no - 08
Roll no - 205

Experiment No - 11:

Name of the experiment:

Potentiometric titration of (a) strong and (b) weak acids by a strong base and their neutralisation curves.

Data:

Data table for the standardization of NaOH against 0.1 N succinic acid:

No. of obs	Volume of succinic acid (mL)	Volume of NaOH solution			Mean (mL)
		IBR (mL)	FBR (mL)	Difference (mL)	
1	10.00	24.0	33.9	9.9	9.9
2	10.00	33.6	43.5	9.9	
3	10.00	26	35.9	9.9	

31.7.23

Data Table-1:

Table for the potentiometric titration of strong acid (HCl) with strong base:

Volume of NaOH (mL)	pH	Volume of NaOH (mL)	pH
0	1.72	10	10.05
1	1.76	11	10.22
2	1.79	12	10.36
3	1.90	13	10.43
4	2.00	14	10.53
5	2.08	15	10.59
6	2.20	16	10.63
7	2.52	17	10.68
8	8.74	18	10.72
9	9.73	19	10.77

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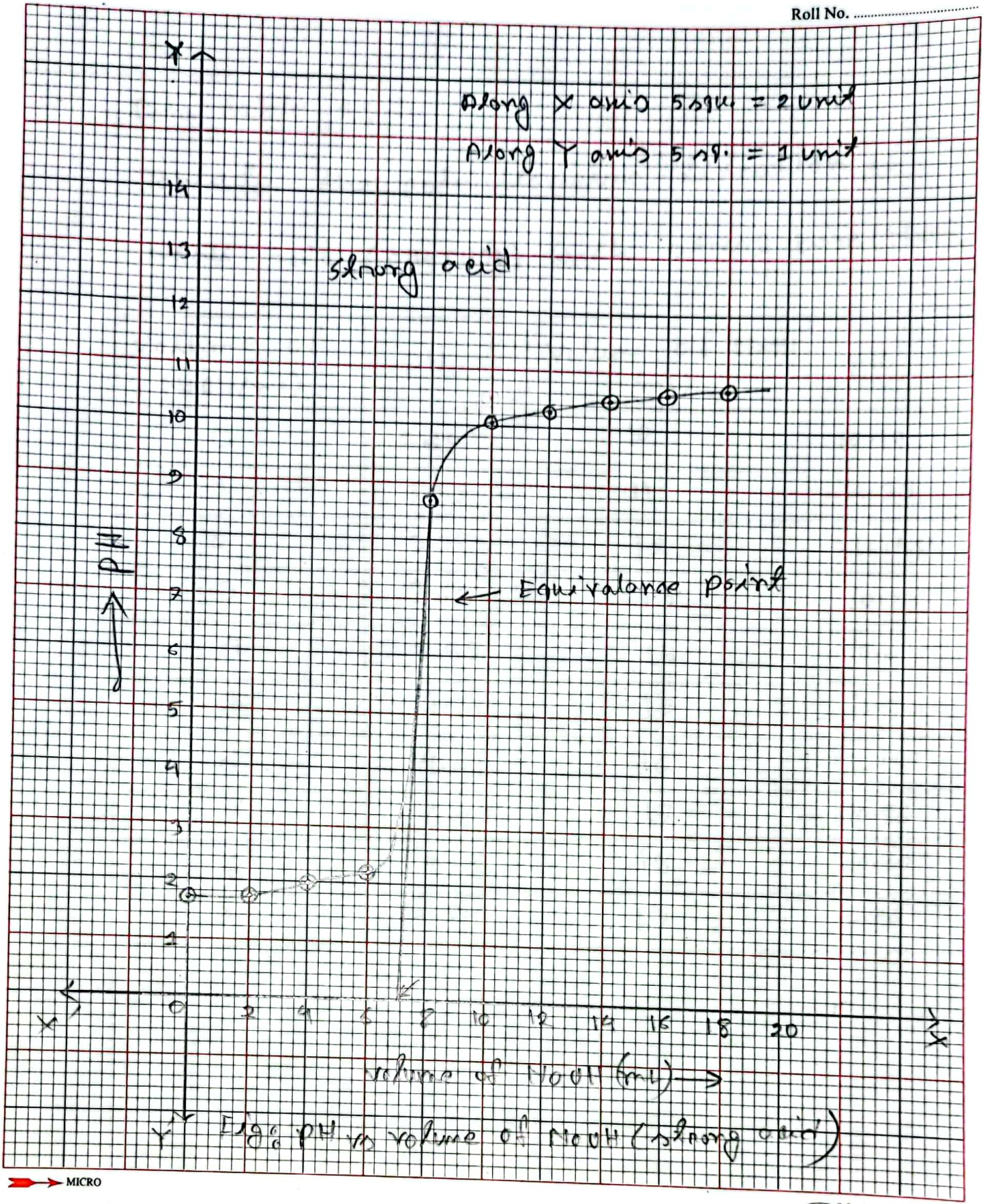

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Data table-2:

Table for the potentiometric titration of weak acid (CH_3COOH) with strong base (NaOH).

Volume of NaOH(mL)	pH	Volume of NaOH (mL)	pH
0	2.68	10	10.10
1	3.03	11	10.33
2	3.35	12	10.48
3	3.63	13	10.58
4	3.89	14	10.67
5	4.14	15	10.74
6	4.41	16	10.80
7	4.82	17	10.86
8	5.58	18	10.90
9	9.54	19	10.96

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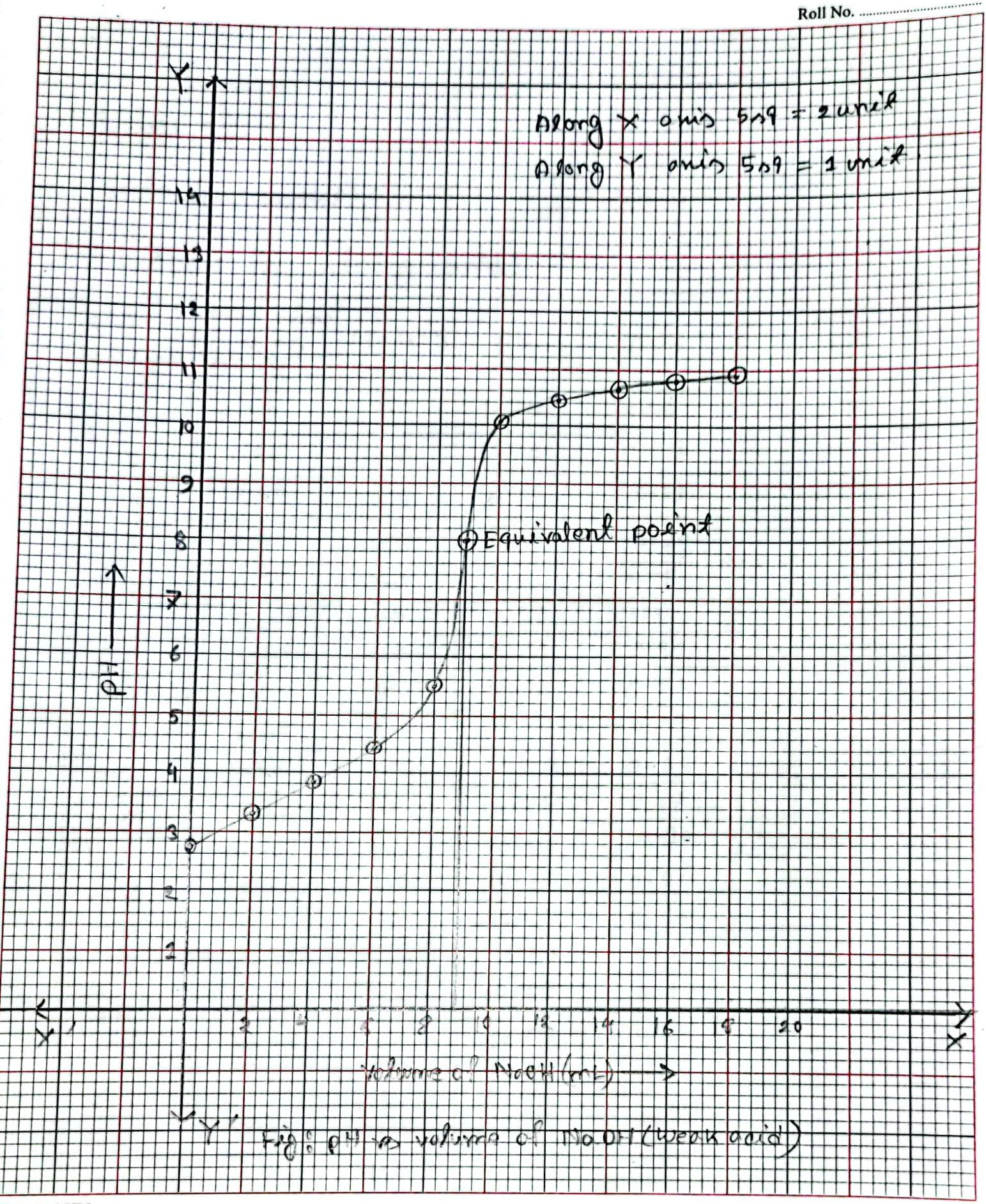

Fig: pH → volume of NaOH (weak acid)

MICRO

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Table-3:

Table for drawing graph $\frac{\Delta pH}{\Delta V}$ vs mean volume of NaOH for strong acid:

Volume of NaOH (mL)	Mean volume $V_m = \frac{V_1 + V_2}{2}$	$\Delta V = V_2 - V_1$ (mL)	pH	Δ pH	$\frac{\Delta pH}{\Delta V}$
0.00	0.50	1.0	1.72	0.04	0.04
1.0	1.50	1.0	1.76	0.03	0.03
2.0	2.50	1.0	1.79	0.11	0.11
3.0	3.50	1.0	1.90	0.10	0.10
4.0	4.50	1.0	2.00	0.08	0.08
5.0	5.50	1.0	2.08	0.12	0.12
6.0	6.50	1.0	2.20	0.32	0.32
7.0	7.50	10	2.52	6.22	6.22
8.0	8.50	1.0	8.74	0.99	0.99
9.0	9.50	1.0	9.73	0.32	0.32
10.0	10.50	1.0	10.05	0.17	0.17
11.0	11.50	1.0	10.22	0.10	0.10
12.0	12.50	1.0	10.36	0.07	0.07
13.0	13.50	1.0	10.43	0.10	0.10
14.0	14.50	1.0	10.53	0.06	0.06
15.0	15.50	1.0	10.59	0.04	0.04
16.0	16.50	1.0	10.63	0.05	0.05
17.0	17.50	1.0	10.68	0.04	0.04
18.0	18.50	1.0	10.72	0.05	0.05
19.0	19.50	1.0	10.77	0.04	0.04

Roll No. .....

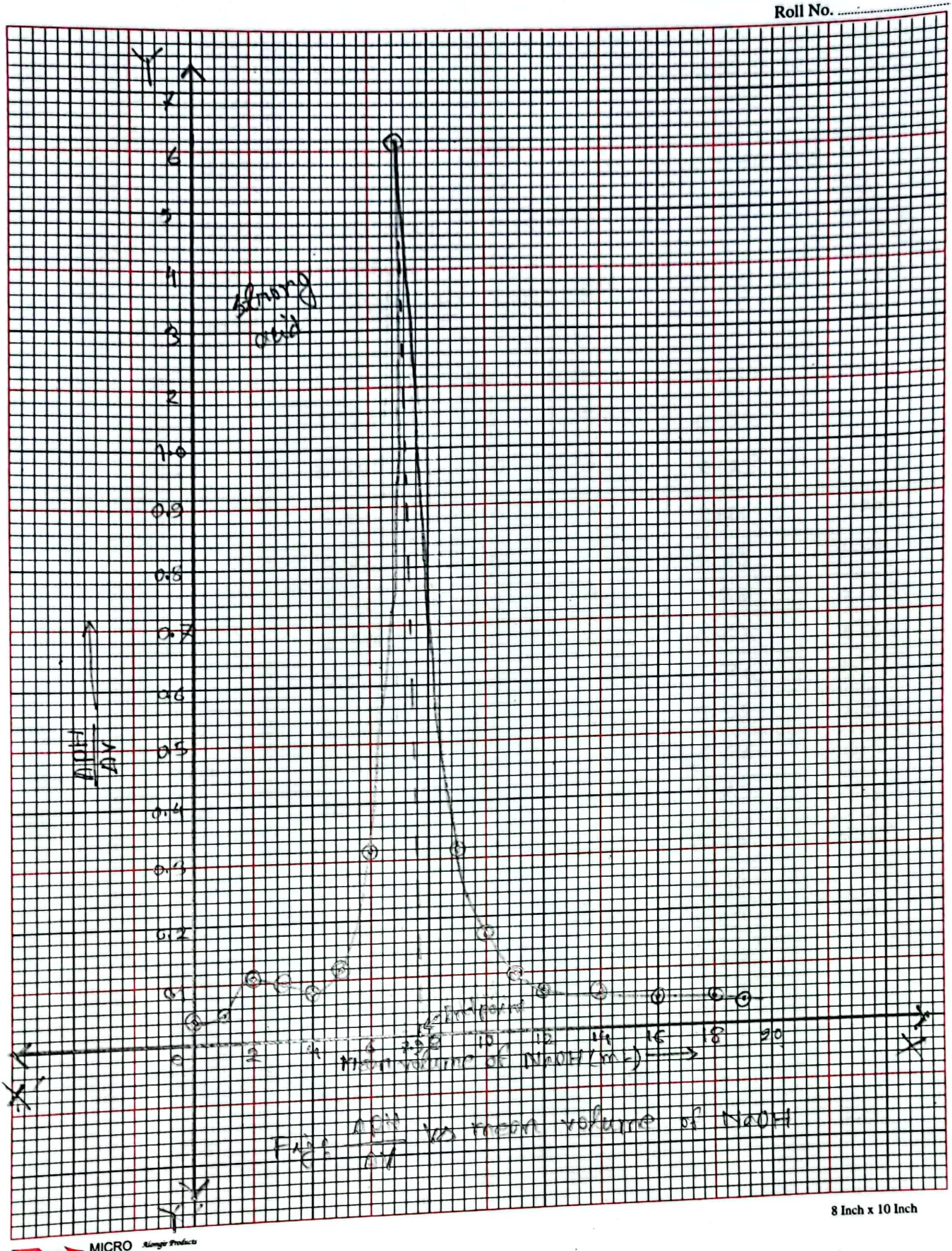

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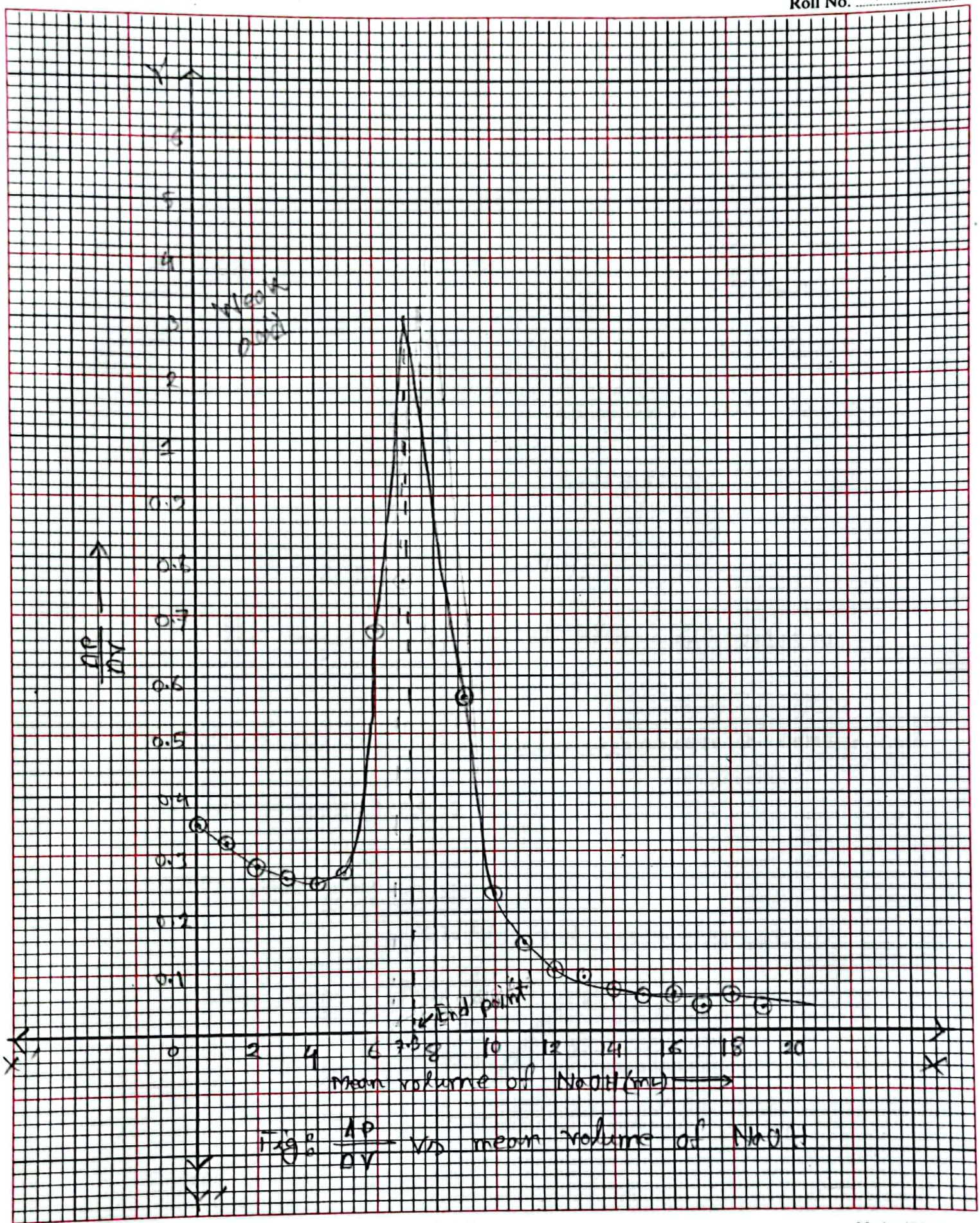
Table-4:

For drawing graph $\frac{\Delta PH}{\Delta V}$ vs mean volume of NaOH of weak acid (CH_3COOH)

volume of NaOH added	Mean volume $V_m = \frac{V_1 + V_2}{2}$	$\Delta V = V_2 - V_1$	pH	$\Delta PH = P_2^H - P_1^H$	$\frac{\Delta PH}{\Delta V}$
0.0	0.50	1.0	2.68	0.35	0.35
1.0	1.50	1.0	3.03	0.32	0.32
2.0	2.50	1.0	3.35	0.28	0.28
3.0	3.50	1.0	3.63	0.26	0.26
4.0	4.50	1.0	3.89	0.25	0.25
5.0	5.50	1.0	4.14	0.27	0.27
6.0	6.50	1.0	4.41	0.68	0.68
7.0	7.50	1.0	4.82	0.76	0.76
8.0	8.50	1.0	5.58	3.93	3.93
9.0	9.50	1.0	9.54	0.56	0.56
10.0	10.50	1.0	10.10	0.23	0.23
11.0	11.50	1.0	10.33	0.15	0.15
12.0	12.50	1.0	10.48	0.10	0.10
13.0	13.50	1.0	10.58	0.09	0.09
14.0	14.50	1.0	10.67	0.07	0.07
15.0	15.50	1.0	10.74	0.06	0.06
16.0	16.50	1.0	10.80	0.06	0.06
17.0	17.50	1.0	10.86	0.04	0.04
18.0	18.50	1.0	10.90	0.06	0.06
19.0	19.50	1.0	10.96	0.06	0.06

Roll No. .....



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$$\Rightarrow \%A = \frac{\qquad}{V_A}$$

Calculations:

calculation of molarity of standard NaOH sol^n:

we know,

$$S_A V_A = S_B V_B$$

$$\Rightarrow S_B = \frac{S_A V_A}{V_B}$$

$$= \frac{10 \times 0.1}{9.9}$$

$$= 0.101 \text{ N}$$

volume of acid, $V_B = 10 mL$

volume of NaOH, $V_B = 9.9 mL$

strength of acid, $S_A = 0.1 N$

strength of NaOH, $S_B = ?$

So, the concentration of NaOH is 0.101 N

(a) calculation for the determination of concentration of strong acid (HCl):

From the graph,

NaOH added, $V_B = 7.9$ mL

volume of HCl, $V_A = 10 mL$

strength of NaOH, $S_B = 0.101 N$

strength of HCl, $S_A = ?$

We know,

$$S_A V_A = S_B V_B$$

$$\Rightarrow S_A = \frac{S_B V_B}{V_A}$$

$$\Rightarrow S_A = \frac{0.101 \times 7.9}{10}$$

$$= 0.079\ N$$

so, the conc. of HCl is 0.079 N

(b) calculation for the determination of the conc. of weak acid (CH_3COOH):

From the graph, NaOH added, $V_B = 7.3$ mL

volume of CH_3COOH, $V_A = 10$ mL

strength of NaOH, $S_B = 0.101$ N

strength of CH_3COOH, $S_A = ?$

we know,

$$S_A V_A = S_B V_B$$

$$\Rightarrow S_A = \frac{0.101 \times 7.3}{10}$$

$$= 0.073\ N$$

so, the strength of CH_3COOH is, 0.073N

Result:

(a) concentration of HCl is 0.079 N

(b) concentration ~~of~~ of CH_3COOH is 0.073 N

Neutralization of HCl and CH_3COOH are shown in graph-1 and graph-2 respectively.

Discussion:

① NaOH is ~~str~~ secondary standard substance, so it should be standardised fast.

② The pH of strong acid and strong base at end point should be near 7.

③ The pH of weak acid and strong base at end point should be greater than 7.

④ The potentiometric titration is more reliable than volumetric titration.

⑤ There may have some error in this experiment.

⑥ We ploted graph carefully.

Date: 01-08-25
Group no - 08
Roll no - 205

Experiment No - 12

Name of the experiment:

Determination of the standard electrode potential of zinc and copper.

Data table - I:

Data for the electrode potential of $ZnSO_4$ soln at different concentration:

Conc. of $ZnSO_4$ solution	potential (V)
0.1	0.954
0.01	0.926
0.001	0.890
0.0001	0.857

Data table - II:

Data for the electrode potential of $CuSO_4$ solution at different concentration:

Concentration of $CuSO_4$ soln	potential (V)
0.1	0.077
0.01	0.060
0.001	0.045
0.0001	0.026

calculation:

Data table for the determination of the standard electrode potential of Zn^{2+} electrode:

concentration of $ZnSO_4$ (M)	EMF of cell (V)	Mean activity of effective γ_+	$M_{Zn^{2+}}\gamma_+$	$\log(M_{Zn^{2+}}\gamma_+)$
0.0001	0.857	0.432	0.0000432	−4.36
0.001	0.890	0.431	0.000431	−3.37
0.01	0.926	0.400	0.00400	−2.40
0.1	0.954	0.161	0.0161	−1.79

Data table for the determination of the standard electrode potential of Cu^{2+} electrode:

concentration of $CuSO_4$ (M)	cell EMF (V)	Mean activity of effective γ_+	$M_{Cu^{2+}}\gamma_+$	$\log(M_{Cu^{2+}}\gamma_+)$
0.0001	0.026	0.437	0.0000437	−4.35
0.001	0.045	0.432	0.000432	−3.36
0.01	0.060	0.404	0.00404	−2.39
0.1	0.077	0.158	0.0158	−1.80

Calculation of $E^{\circ}_{Zn/Zn^{2+}}$:

The cell in ⇄ represent as,

$$Zn, Zn^{2+}_{(aq)} \,||\, \text{calomel (saturated)}$$

The emf of the cell, $E_{cell} = E^{\circ}_{Zn/Zn^{2+}} + E_{calomel\,(red)}$

$$\therefore E_{Zn/Zn^{2+}} = E_{cell} + \frac{2.303\,RT}{2F} \log\left(M_{Zn^{2+}} \cdot \gamma_{\pm}\right) - E_{cell\,(red)}$$

Redox potential of saturated standard calomel electrode = 0.244 V

(1) For 0.0001 M $ZnSO_4$ solution:

$$E_{Zn/Zn^{2+}} = 0.857 + \frac{2.303 \times 8.314 \times 300}{2 \times 96500} \times (4.36) - 0.244$$

$$= 0.48\ V$$

(2) For 0.001 M $ZnSO_4$ solution:

$$E_{Zn/Zn^{2+}} = 0.890 + \frac{2.303 \times 8.314 \times 300}{2 \times 96500} \times (-3.37) - 0.244$$

$$= 0.54\ V$$

③ For 0.01 M $ZnSO_4$ solution:

$$E_{Zn/Zn^{2+}} = 0.926 + \frac{2.303 \times 8.314 \times 300}{2 \times 96500} \times (-2.40) - 0.244$$

$$= 0.61 \text{ V}$$

④ For 0.1 M $ZnSO_4$ solution:

$$E_{Zn/Zn^{2+}} = 0.954 + \frac{2.303 \times 8.314 \times 300}{2 \times 96500} \times (-1.79) - 0.244$$

$$= 0.656 \text{ V}$$

Calculation of $E^\circ_{Cu^{2+}/Cu}$:

The cell is represent as,

$$\text{calomel (sat)} \,||\, Cu^{2+}_{(aq)}, Cu$$

The emf of the cell, $E_{cell} = E_{cal(sat)(ox)} + E_{Cu^{2+}/Cu}$

$$= E_{cal(ox)} - E_{Cu/Cu^{2+}}$$

$$= E_{cal(ox)} - E^\circ_{Cu/Cu^{2+}} + \frac{2.303RT}{2F} \log(M_{Cu^{2+}} \gamma_+)$$

$$\therefore E_{cell} = -E_{cal(red)} + E^\circ_{Cu^{2+}/Cu} + \frac{2.303RT}{2F} \log(M_{Cu^{2+}} \cdot \gamma_+)$$

$$\therefore E^\circ_{Cu^{2+}/Cu} = E_{cell} - \frac{2.303RT}{2F} \log(M_{Cu^{2+}} \cdot \gamma_+) + E_{cal(red)}$$

① For 0.0001 M $CuSO_4$ solution:

$$E_{Cu^{2+}/Cu} = 0.026 - \frac{2.303 \times 8.314 \times 300}{2 \times 96500} \times (-4.35) + 0.244$$

$$= 0.399\ V$$

② For 0.001 M $CuSO_4$ solution:

$$E_{Cu^{2+}/Cu} = 0.045 - \frac{2.303 \times 8.314 \times 300}{2 \times 96500} \times (-3.36) + 0.244$$

$$= 0.389\ V$$

③ For 0.01 M $CuSO_4$ solution:

$$E_{Cu^{2+}/Cu} = 0.060 - \frac{2.303 \times 8.314 \times 300}{2 \times 96500} \times (-2.39) + 0.244$$

$$= 0.375\ V$$

④ For 0.1 M $CuSO_4$ solution:

$$E_{Cu^{2+}/Cu} = 0.077 - \frac{2.303 \times 8.314 \times 300}{2 \times 96500} \times (-1.80) + 0.244$$

$$= 0.374\ V$$

Table: $\log M_{Zn^{2+}}$ vs $E^{\circ}_{Zn/Zn^{2+}}$ for $ZnSO_4$ soln:

$\log M_{Zn^{2+}}$	$E^{\circ}_{Zn/Zn^{2+}}$ (V)
-4.36	0.48
-3.37	0.54
-2.40	0.61
-1.79	0.656

Table: $\log M_{Cu^{2+}}$ vs $E^{\circ}_{Cu^{2+}/Cu}$ for $CuSO_4$ soln:

$\log M_{Cu^{2+}}$	$E^{\circ}_{Cu^{2+}/Cu}$ (V)
-4.35	0.399
-3.36	0.389
-2.39	0.375
-1.80	0.374

From the graph-1 the standard electrode potential of Zn is 0.76 V (oxidation potential)
The standard potential of Cu is 0.348 V (reduction potential)

Result:

① The standard electrode potential of Zn is $Zn/Zn^{2+} = 0.76V$ (oxidation potential)

② The standard electrode potential of Cu is $Cu^{2+}/Cu = 0.345V$ (reduction potential)

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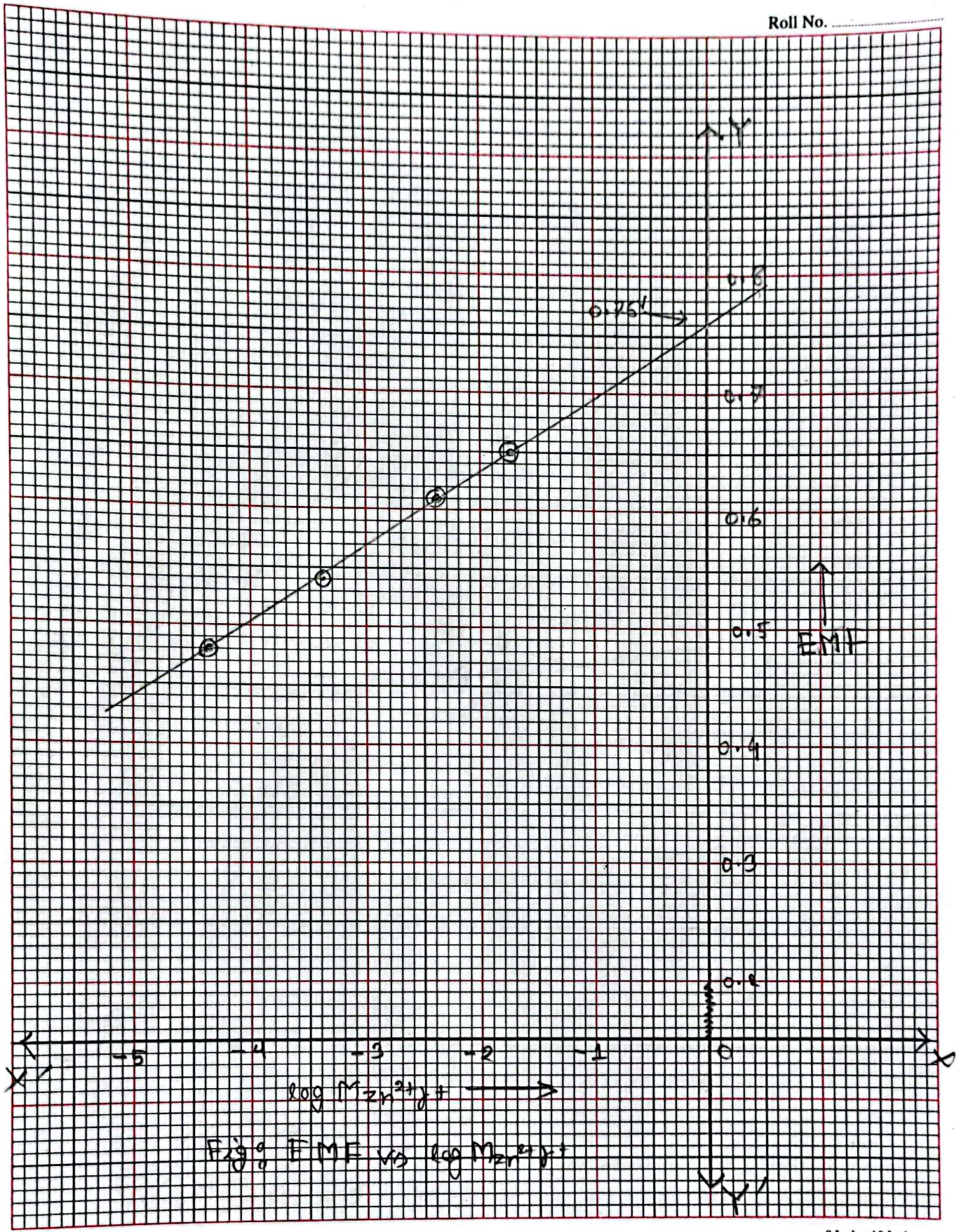

Fig: EMF vs log $M_{Zn^{2+}}$

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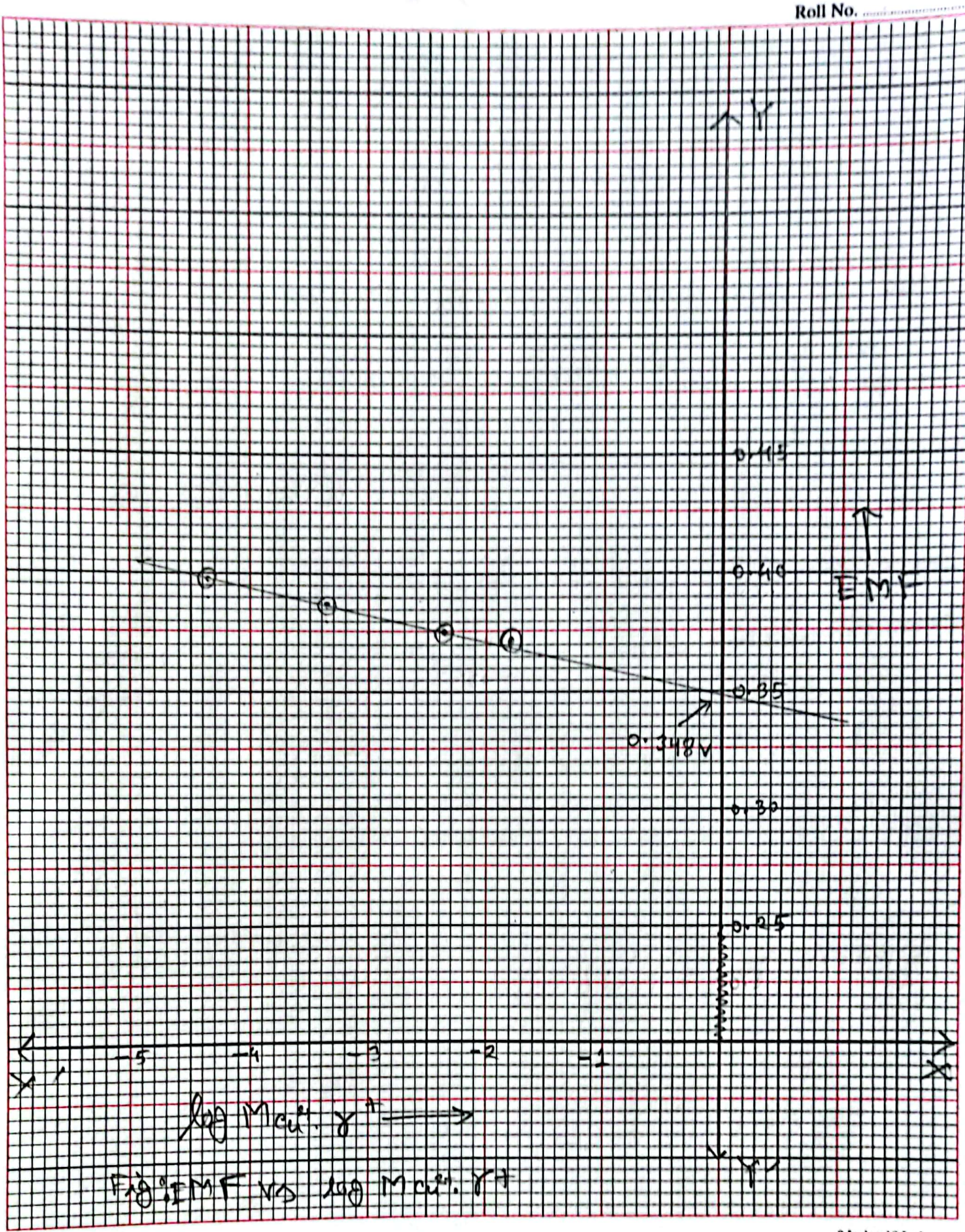

Fig: EMF vs log Mcu². γ+

Discussion:

① Solution of different concentration prepared carefully.

② Salt bridge has washed during experiment with different concentration of solution.

③ potentiometer reading has taken carefully.

④ E.M.F of cell using different concentration of $CuSO_4$ and $ZnSO_4$ measured carefully.

⑤ The graph was plotted preciously as much as possible.